

PACKING $k^2 - c$ UNIT SQUARES: $s(k^2 - c) = k$ **FOR ALL LARGE k**

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ABSTRACT. Let $s(n)$ be the side of the smallest square into which n non-overlapping unit squares can be packed. We show that for every fixed $c \geq 0$ one has $s(k^2 - c) = k$ for all sufficiently large integers k . Equivalently, if $M(k)$ denotes the largest number of unit squares that can be placed in $[0, k]^2$ so that they are pairwise disjoint as closed sets, then $k^2 - M(k) \rightarrow \infty$; in fact $k^2 - M(k) \geq \kappa \log k$ for every $k \geq 2$, with an absolute constant $\kappa > 0$. The proof combines a sharpened form of the fundamental lemma of Roth and Vaughan, with an explicit constant and a further sharpening for nearly parallel pairs, with two simple observations: a closedness constraint on horizontal lines, which is special to integer sides, and a quantization of heights in columns of nearly axis-parallel squares, measured from the bottom and from the top side and organised over a continuum of scales; a count of the cracks that can break the quantization controls the columns, and an angular argument, sharpened by a computer-assisted branch-and-bound computation, bounds the overlap of the Roth–Vaughan rectangles at uncovered points. We obtain $\kappa = 0.0353$, and $k^2 - M(k) \geq (0.0657 - o(1)) \log k$ as $k \rightarrow \infty$. Since $k^2 - M(k) \geq 1$ for every $k \geq 2$ by an elementary argument, the logarithmic bound has content only for $\log k > 1/\kappa \approx 28$, and the result should be regarded as qualitative. With Daniel’s unrefereed computer-assisted result $s(k^2 - 3) = k$ ($k \geq 6$), $\kappa = 0.0541$.

1. INTRODUCTION

For $n \geq 1$ let $s(n)$ denote the side of the smallest square that contains n unit squares with pairwise disjoint interiors (rotations allowed). Trivially $s(k^2) = k$, and $s(n) \leq \lceil \sqrt{n} \rceil$. For $n = k^2 - c$ with c small compared with k it is natural to expect that one cannot do better than the trivial $k \times k$ grid with c squares removed. Nagamochi [23] stated $s(k^2 - 1) = s(k^2 - 2) = k$ for all $k \geq 2$; the scoring assertion of his Lemma 1, on which the published proof rests, has recently been shown to fail [7, 18] (Karakuş points out that this does not disprove the rectangle bound that Nagamochi derives from it), and replacement proofs have been given: of $s(k^2 - 1) = k$ in [18], through a weaker rectangle bound proved with a strip measure, and of $s(k^2 - 2) = k$ (which implies $s(k^2 - 1) = k$), with a formalization in Lean, in [7]; see also the discussion in [10]. For $k^2 - 3$ the values $k = 3, \dots, 7$ are known [19, 3, 4] ($k = 5, 6$ in the preprint [4]), and a computer-assisted proof of $s(k^2 - 3) = k$ for all $k \geq 6$, not yet refereed, has recently been announced [10] (see Corollary 1.3 and the paragraph after it); the same repository has since announced, again by a computer-assisted and unrefereed argument, that $s(k^2 - 4) = k$ for all $k \geq 5$ [10, directory `s12/certificates/k2m4`]. Friedman’s survey [15] contains the conjecture that $s(n^2 - c) = n$ implies $s((n + 1)^2 - c) = n + 1$, and Chung and Graham [8, concluding remarks] wrote that it seems likely that $s(n^2 - 1) = n$, and that the same should hold for $n^2 - c$ unit squares when c is fixed and n is sufficiently large (attributing this to [15]). On the other hand, $s(k^2 - c) < k$ as soon as c is large in terms of k : Arslanov, Mustafin and Shangitbayev [2] show $s(n^2 - n) < n$ for $n \geq 12$. Erdős and Graham [11] packed a square of side x with wasted area $O(x^{7/11})$, and this exponent was lowered several times (see the introductions of [22, 6]); the exponent $3/5$ was claimed by Chung and Graham [9]; an error in their argument was pointed

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out by Arslanov and Bui [1], and the bound $O(x^{3/5})$ was re-established, independently and by different methods, in the preprints of Bui [6] and of McClenagan [22] (the latter based on his 2024 thesis). Applied to x slightly below k , this gives $s(k^2 - Ck^{3/5}) < k$ for large k .

Following the notes [10, file `s12/search/FRIEDMAN.md`], define

$$c^*(k) := \max\{c \in \mathbb{Z}_{\geq 0} : s(k^2 - c) = k\}.$$

Since s is non-decreasing, $s(k^2 - c) = k$ holds exactly for $0 \leq c \leq c^*(k)$, and Friedman's conjecture says precisely that c^* is non-decreasing. The results above give $3 \leq c^*(k) \leq O(k^{3/5})$ for $k \geq 3$; here the lower bound uses the unrefereed announcement [10] for $k \geq 8$ (and the later announcement would give $c^*(k) \geq 4$ for $k \geq 5$), and the upper bound uses the preprints [22, 6] (the refereed bound $O(x^{(3+\sqrt{2})/7} \log x)$ of [8] gives $c^*(k) = O(k^{(3+\sqrt{2})/7} \log k)$). Whether $c^*(k) \rightarrow \infty$, i.e. whether for every fixed c one has $s(k^2 - c) = k$ for all large k , as expected in [8], is recorded as open in [10], where it is also observed that it does not follow from the lower bound of Roth and Vaughan [25]: their theorem [25, p. 170, Theorem] states that the wasted area $w(\alpha)$ in a square of side α satisfies $w(\alpha) \gg (\alpha \|\alpha\|)^{1/2}$ whenever $\alpha(\alpha - [\alpha]) > 1/6$, where $\|\alpha\|$ is the distance from α to the nearest integer, and this vanishes as $\alpha \rightarrow k^-$. We are not aware of a proof in the literature that $c^*(k)$ is unbounded.

Theorem 1.1. *There is an absolute constant $\kappa > 0$ such that for every integer $k \geq 2$, every family of N unit squares in $[0, k]^2$ that are pairwise disjoint as closed sets satisfies*

$$k^2 - N \geq \kappa \log k.$$

One may take $\kappa = 0.0353$ ($\kappa = 0.0319$ without the computer-assisted Lemma 4.10, see Remark 7.5; both constants of this theorem use that lemma). Moreover, for every $k \geq 2$ one has $k^2 - N \geq 1.99954(\log k - 13.06675)/30.418$, so that $k^2 - N \geq (0.0657 - o(1)) \log k$ as $k \rightarrow \infty$. (Here and below $\log$ is the natural logarithm.)

Corollary 1.2. *$c^*(k) \geq \kappa \log k - 1$ for $k \geq 2$. In particular $c^*(k) \rightarrow \infty$: for every fixed c , $s(k^2 - c) = k$ for all sufficiently large k . This confirms the expectation of Chung and Graham [8] stated above.*

Corollary 1.3 (using an unrefereed computer-assisted result). *Assume that $s(k^2 - 3) = k$ for every integer $k \geq 6$, as announced, with a computer-assisted proof that has not been refereed, in [10, directory `s12/certificates/k2m3`]. Then every family of N unit squares in $[0, k]^2$ that are pairwise disjoint as closed sets satisfies $k^2 - N \geq 4$ if $k \geq 6$, and*

$$k^2 - N \geq 0.0541 \log k \quad \text{for every integer } k \geq 2.$$

Proof. Let $k \geq 6$. Since $s(k^2 - 3) = k$ is not less than k , Lemma 2.1 gives $k^2 - 3 > M(k)$, so $k^2 - N \geq k^2 - M(k) \geq 4$. Let $F(k) := 1.99954(\log k - 13.06675)/30.418$ be the second bound of Theorem 1.1, and $L_4 := 13.06675 + 4 \cdot 30.418/1.99954 = 73.91674\dots$, so that $F(k) \geq 4$ exactly when $\log k \geq L_4$. For $2 \leq k \leq 5$, Corollary 3.3 gives $k^2 - N \geq 1 > 0.0541 \log 5 \geq 0.0541 \log k$. For $k \geq 6$ with $\log k \leq L_4$ we have $0.0541 \log k \leq 0.0541 L_4 < 3.999 < 4 \leq k^2 - N$. For $\log k > L_4$, Theorem 1.1 gives $k^2 - N \geq F(k)$, and $F(k)/\log k = \frac{1.99954}{30.418}(1 - \frac{13.06675}{\log k})$ increases with $\log k$ and equals $4/L_4 > 0.05411$ at $\log k = L_4$; hence $k^2 - N \geq F(k) > 0.05411 \log k$. $\square$

The assumption of Corollary 1.3 concerns all $k \geq 6$ at once. In [10] it is derived, for each such k , from a measure on $[0, k]^2$ of total mass $k^2 - 4D$, $D = 423621306389/500000000000 > 3/4$, under which every closed unit square in $[0, k]^2$ has mass at least 1; this property is reduced, for all $k \geq 6$ simultaneously, to one finite statement about $[0, 7]^2$. The reduction and the axis-parallel case of the finite statement are checked in Lean 4 (theorems `bentz_of_valid7`, `validAxis7` and `bentz_of_validTilt7` in [10]), while the case of tilted squares is certified by an exact computer

search in rational arithmetic and is not formalized. The finite statement was certified again by an independently written exact checker [26], and the Squares Project [20, entry T-064] replayed both checkers in full and rebuilt the Lean reduction. There is no refereed publication of this result, and the recorded reviews of it are by AI agents; the repository states that the work was produced by Claude (Anthropic) under human direction. Theorem 1.1 does not use [10]; Corollary 1.3 changes only the constant for all k , not the asymptotic slope 0.0657. The assumption is needed only for $6 \leq k \leq e^{58.71}$: for $\log k > 13.06675 + 3 \cdot 30.418/1.99954 = 58.7042 \dots$ the second bound of Theorem 1.1 exceeds 3, and W is an integer. Through F , Corollary 1.3 also uses the computer-assisted Lemma 4.10; with $K_w = 13$ (Remark 7.5) the same argument gives $4/86.0695 \dots > 0.0464$.

Since $k^2 - N \geq 1$ for every $k \geq 2$ (Corollary 3.3), the bound $\kappa \log k$ says something new only for $\log k > 1/\kappa \approx 28.3$; the second bound of the theorem exceeds 1 only for $\log k > 28.279$, so for smaller k the theorem is just Corollary 3.3 (see Remark 7.3).

Idea of the proof. On a horizontal line the squares cut out pairwise disjoint closed intervals of $[0, k]$, so at most $k - 1$ of them have length ≥ 1 (Lemma 3.2). A square of inclination a is cut in an interval of length $\sec a \geq 1$ except near its top and bottom (“ramps”, of height $\sin a$). Hence, unless the line carries some waste, it must either cross a square of large inclination, which by a sharpened form of the Roth–Vaughan fundamental lemma (Proposition 4.3) costs waste proportional to the inclination, or cross a nearly axis-parallel square through its ramp. In the latter case we look down the column below that square: if the column is “rigid” (little vertical waste, only tiny inclinations) and avoids the narrow “ends” of the squares in it, heights in it are quantized, i.e. the square’s corners sit within $1.265 \cdot 10^{-5}$ of integers, so its ramp cannot meet a line at a height whose fractional part is farther than that from 0 and 1. A rigid column through an end of a square must pass through a tight crack between two squares, and at most one crack per unit of height fits under a square; so the exceptional columns have measure about $1/C$, where $1/(Cs)$ is the inclination threshold for columns at scale s . Thus many columns must be non-rigid, which again costs waste. Since k is an integer, the same holds for columns hanging from the top side. A line at height y is treated with the inclination threshold $1/(C'd(y))$, where $d(y)$ is the distance to the nearer of the bottom and the top side, and each column is charged only at the least scale at which it fails to be rigid; integrating over heights y with $y_{\min} \leq d(y) \lesssim k/2$, where $y_{\min} = 236000$ is an absolute constant, with the measure $dy/d(y)$ then gives a lower bound of order $\log k$; within this scheme the choice $C' = 2\sqrt{2}$, $C = 2$ is optimal. Lines this close to the walls can be used because the excess over 1 of the long chords on a line is itself charged to the inclinations of the squares in the columns between them and the nearer side (Lemma 4.15), at a cost whose integral against $dy/d(y)$ is bounded. Further refinements matter for the constant: on a line with little waste the short chords cut from ramps add up to almost a full unit, for nearly parallel, nearly touching squares the fundamental lemma holds with a much better constant (Proposition 4.5), and only the uncovered points of the Roth–Vaughan rectangles matter, where at most 13 of them can overlap (Lemma 4.9), and at most 9 by a computer-assisted refinement (Lemma 4.10). Only the numerical constants ($\kappa = 0.0353$ and the explicit bound) depend on the computer: with Lemma 4.9 alone the proof gives $\kappa = 0.0319$ (Remark 7.5).

2. CLOSED PACKINGS AND THE REDUCTION

Throughout, a *square* is a closed unit square. A *closed packing* of $[0, k]^2$ is a finite family of squares contained in $[0, k]^2$ that are pairwise disjoint as closed sets. Put $M(k)$ for the maximum size of a closed packing of $[0, k]^2$ and, for a given closed packing, $W := k^2 - N$, the area of $[0, k]^2$ not covered by the squares.

Lemma 2.1. *For integers $k \geq 2$ and $n \geq 1$: $s(n) < k$ if and only if $n \leq M(k)$. Moreover $M(k) \leq k^2 - 1$ (Corollary 3.3 below), so $c^*(k) = k^2 - M(k) - 1$.*

Proof. The minimum in the definition of $s(n)$ is attained by compactness. If n squares with disjoint interiors lie in a square of side $s < k$, place that square concentrically in $[0, k]^2$, let O be the common centre, and replace each centre c_i by $O + \lambda(c_i - O)$ with $\lambda = k/s > 1$, keeping orientations. If two of the original squares are separated by a line with normal u , the separation $u \cdot (c_2 - c_1) \geq h_1(u) + h_2(u)$ (where $h_i(u)$ is the half-width of the i -th square in direction u) becomes strict after dilation, and every moved square stays in $[0, k]^2$: in coordinates centred at O , $|c_i \cdot e| + h_i(e) \leq s/2$ for e a coordinate direction implies $\lambda|c_i \cdot e| + h_i(e) \leq \lambda s/2 - (\lambda - 1)h_i(e) \leq k/2$. So we obtain a closed packing of $[0, k]^2$ with n squares. Conversely, a closed packing has a positive minimum distance η between its squares; contracting all centres about the centre of $[0, k]^2$ by a factor $\mu < 1$ close to 1 keeps the squares disjoint and places them in a concentric square of side $\mu k + (1 - \mu)\sqrt{2} < k$. Finally, $s(n) \leq k$ always holds for $n \leq k^2$, so for integers $0 \leq c < k^2$ we have $s(k^2 - c) = k$ iff $k^2 - c > M(k)$. $\square$

Thus Theorem 1.1 implies Corollary 1.2. Note that W is an integer.

3. ELEMENTARY GEOMETRY

The *inclination* $a(S) \in [0, \pi/4]$ of a square S is the least angle between a side of S and a side of $[0, k]^2$; the inclination $\theta(S_1, S_2)$ between two squares is the least angle between their sides (this is the quotient metric on $\mathbb{R}/(\pi/2)\mathbb{Z}$, with values in $[0, \pi/4]$), and we write $S_0 := \partial[0, k]^2$ with $\theta(S, S_0) := a(S)$. These are the definitions of [25]. Likewise, for a single side σ of $[0, k]^2$ we put $\theta(S, \sigma) = \theta(\sigma, S) := a(S)$.

Lemma 3.1. *Let S be a square with inclination a and lowest point at height v .*

- (a) *The vertical extent of S is $[v, v + \cos a + \sin a]$.*
- (b) *For $y \in \mathcal{M}(S) := [v + \sin a, v + \cos a]$ (the middle band, of length $\cos a - \sin a$) the line $y = \text{const}$ meets S in a segment of length $\sec a \geq 1$; for y in the ramp set $\mathcal{R}(S) := [v, v + \sin a] \cup (v + \cos a, v + \cos a + \sin a]$ the length is $< \sec a$. If $a = 0$, then $\mathcal{R}(S) = \emptyset$.*
- (c) *Symmetrically, for x in the vertical middle band $\mathcal{M}'(S)$, an interval of length $\cos a - \sin a$ (defined in the proof; for $a = 0$ it is the horizontal extent of S), the vertical line through x meets S in a segment of length $\sec a$, entering S through a side of slope $\pm \tan a$; the lower endpoint of that segment has height in $[v, v + \sin a]$. The complement of $\mathcal{M}'(S)$ in the horizontal extent of S (the vertical ramp set) has total length $2 \sin a$.*

Proof. For $a = 0$ everything is clear. Let $0 < a \leq \pi/4$ and (up to reflection) let the vertices be $P_0 = (x_0, v)$, $P_1 = P_0 + (\cos a, \sin a)$, $P_3 = P_0 + (-\sin a, \cos a)$, $P_2 = P_1 + P_3 - P_0$, at heights $v, v + \sin a, v + \cos a, v + \cos a + \sin a$. For $y \in [v + \sin a, v + \cos a]$ the left and right boundary points lie on the parallel sides P_0P_3 and P_1P_2 , which are at distance 1 and have direction $(-\sin a, \cos a)$; hence the horizontal chord is $1/\cos a$. Outside this band one end of the chord lies on P_0P_1 or P_3P_2 , and the chord is shorter. For (c), put $\mathcal{M}'(S) := [x_0, x_0 + \cos a - \sin a]$ (in the reflected position, its mirror image $[x_0 - \cos a + \sin a, x_0]$). On this interval the lower and upper boundary points lie on P_0P_1 and P_3P_2 , parallel sides at distance 1 with direction $(\cos a, \sin a)$, so the vertical chord is $\sec a$; the lower point has height $v + (x - x_0) \tan a \in [v, v + (\cos a - \sin a) \tan a] \subset [v, v + \sin a]$. The horizontal extent has length $\cos a + \sin a$, so the vertical ramp set has length $2 \sin a$. $\square$

For $y \in [0, k]$ let $\omega(y)$ be the length of the part of the segment $[0, k] \times \{y\}$ not covered by squares (the *line waste*); by Fubini $\int_0^k \omega(y) dy = W$. Similarly, for a vertical segment σ we call the length of σ not covered by squares its *vertical waste*; integrating the vertical waste of

$\{x\} \times [0, k]$ over x also gives W . Since a packing is finite, all functions of y (or of x) considered below are piecewise continuous with finitely many pieces, and all sets of y (or of x) that we define are finite unions of intervals; we shall not comment on measurability again.

Lemma 3.2 (closedness). *Let $k \geq 2$. On every horizontal line, the squares of a closed packing of $[0, k]^2$ meet the segment $[0, k] \times \{y\}$ in pairwise disjoint closed intervals, and at most $k - 1$ of these have length ≥ 1 .*

Proof. Disjointness is inherited from the squares. If k of the intervals had length ≥ 1 , their union would be a closed subset of $[0, k]$ of full measure, hence all of $[0, k]$, which is then a disjoint union of $k \geq 2$ nonempty closed sets, contradicting connectedness. $\square$

Corollary 3.3. *For $k \geq 2$ every closed packing of $[0, k]^2$ has $W \geq 1$; that is, $M(k) \leq k^2 - 1$.*

Proof. W is an integer. If $W = 0$, then $\omega(y) = 0$ for some (indeed almost every) y , so the intervals cut out on that line cover $[0, k]$ up to a null set; each has length $\leq \sqrt{2} < k$, so there are at least two, and as in the proof of Lemma 3.2 their union is a closed set of full measure, hence $[0, k]$, a contradiction. $\square$

For a square S and $y \in \mathbb{R}$ let $c_S(y)$ be the length of the horizontal chord $S \cap (\mathbb{R} \times \{y\})$ (zero if the line misses S).

Lemma 3.4 (short chords). *Let $k \geq 2$, $y \in (0, k)$ and $0 < \alpha \leq \frac{\pi}{4}$, and suppose that every square meeting the line y has inclination $< \alpha$. Then*

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) > 1 - \omega(y) - (k - 1)(\sec \alpha - 1),$$

the sum being over the squares S of the packing. Moreover, for a square S let $\Phi(S) := \{y : 0 < c_S(y) < 1\}$. If $a := a(S) > 0$, then $\Phi(S) \subset \mathcal{R}(S)$, $|\Phi(S)| = 2 \sin a \cos a$ and $\int_{\Phi(S)} c_S(y) dy = \sin a \cos a$; if $a(S) = 0$, then $\Phi(S) = \emptyset$.

Proof. By Lemma 3.1(b), a chord of length ≥ 1 has length at most $\sec a(S) < \sec \alpha$, and by Lemma 3.2 there are at most $k - 1$ such chords on the line; the chords cover a length $k - \omega(y)$ of $[0, k]$, which gives the inequality. For the second part let $a > 0$ and let v be the height of the lowest point of S ; in the coordinates of the proof of Lemma 3.1, for $v \leq y \leq v + \sin a$ the chord joins the sides P_0P_3 and P_0P_1 and has length $(y - v)(\tan a + \cot a) = (y - v)/(\sin a \cos a)$, which is < 1 exactly for $y < v + \sin a \cos a$; symmetrically at the top, and in the middle band the chord is $\sec a \geq 1$. Hence $\Phi(S)$ consists of two intervals of length $\sin a \cos a$ in $\mathcal{R}(S)$, on each of which c_S grows linearly from 0 to 1, so its integral there is $\frac{1}{2} \sin a \cos a$. If $a = 0$, every nonzero chord has length 1. $\square$

Without an assumption on the inclinations, the same counting gives a bound in terms of the excess of the long chords.

Lemma 3.5 (short chords and the excess of long chords). *Let $k \geq 2$ and $y \in (0, k)$, and put*

$$E(y) := \sum_{S: c_S(y) \geq 1} (c_S(y) - 1),$$

the sum being over the squares S of the packing. Then

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) \geq 1 - \omega(y) - E(y).$$

Proof. The chords $S \cap (\mathbb{R} \times \{y\})$ lie in $[0, k] \times \{y\}$, are pairwise disjoint, and cover a length $k - \omega(y)$ of it. By Lemma 3.2, the number n_L of squares with $c_S(y) \geq 1$ is at most $k - 1$, and these chords have total length $n_L + E(y) \leq k - 1 + E(y)$. The remaining chords therefore have total length at least $k - \omega(y) - (k - 1) - E(y)$. $\square$

Under the assumption of Lemma 3.4, $E(y) < (k-1)(\sec \alpha - 1)$ by Lemma 3.1(b), so Lemma 3.5 contains the inequality of Lemma 3.4; Lemma 4.15 below bounds $E(y)$ instead by the waste.

4. A SHARPENED FUNDAMENTAL LEMMA AND THE COST OF INCLINATIONS

Throughout this section $k \geq 4$, a closed packing of $[0, k]^2$ is fixed, $S_0 := \partial[0, k]^2$, and the *waste* is the set of points of $(0, k)^2$ that lie in no square; its area is W . (In the proof of Theorem 1.1 this section and the next are used only for $k \geq 10^{12}$; for $2 \leq k < 10^{12}$ the theorem follows from Corollary 3.3, see Section 6.) We begin with two elementary facts about gaps between convex sets.

Lemma 4.1 (convex gaps). *Let I be an interval of length λ , let $0 \leq \varphi \leq 1$, and let $g \geq 0$ be a continuous, convex, piecewise linear function on I with finitely many break points, such that $|g'| \geq \varphi$ off the break points and the slope of g increases by at least 2 at every break point. Then $\int_I g \geq \varphi(2 - \varphi)\lambda^2/4$.*

Proof. We may assume $\varphi > 0$ and $\lambda > 0$. Let y^* be a minimum point of g on the closure of I , and let a and $b = \lambda - a$ be the lengths of the parts of I to the left and to the right of y^* . If y^* is an end point of I , then g is monotone with $|g'| \geq \varphi$, so $g(y) \geq \varphi|y - y^*|$ and $\int_I g \geq \varphi\lambda^2/2 \geq \varphi(2 - \varphi)\lambda^2/4$. Otherwise y^* is a break point, since $g' \neq 0$ elsewhere; let $-p \leq 0 \leq q$ be the slopes of g just to the left and just to the right of y^* . Then $p, q \geq \varphi$ and $p + q \geq 2$, and by convexity $g(y) \geq p(y^* - y)$ for $y < y^*$ and $g(y) \geq q(y - y^*)$ for $y > y^*$. By the Cauchy–Schwarz inequality, $\lambda^2 = (a + b)^2 \leq (pa^2 + qb^2)(1/p + 1/q)$, so

$$\int_I g \geq \frac{pa^2 + qb^2}{2} \geq \frac{\lambda^2}{2(1/p + 1/q)}.$$

Let, say, $p \leq q$. If $q \geq 2 - \varphi$, then $1/p + 1/q \leq 1/\varphi + 1/(2 - \varphi)$. If $q < 2 - \varphi$, then $q \geq 1$ (as $p + q \geq 2$) and $p \geq 2 - q$, so $1/p + 1/q \leq F(q) := 1/(2 - q) + 1/q$; the function F is convex on $(0, 2)$ and symmetric about 1, so $F(q) \leq F(2 - \varphi) = 1/\varphi + 1/(2 - \varphi)$. In both cases $1/p + 1/q \leq 2/(\varphi(2 - \varphi))$, and the lemma follows. $\square$

Let (s, t) be Cartesian coordinates in the plane. For a square Z let $[\tau^-(Z), \tau^+(Z)]$ be its projection to the t -axis, and for $\tau^-(Z) \leq t \leq \tau^+(Z)$ let $l_Z(t)$ and $r_Z(t)$ be the least and the largest s with $(s, t) \in Z$. The vertices of Z at which $t = \tau^+(Z)$ or $t = \tau^-(Z)$ are its t -extreme vertices.

Lemma 4.2 (boundary functions). *The function l_Z is convex and r_Z is concave. Each of them is piecewise linear with at most one break point, at which the slope of l_Z increases, and that of r_Z decreases, by at least 2. Off the break points $l'_Z(t) = \tan \psi$, where $\psi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is the angle between the t -axis and the side of Z containing $(l_Z(t), t)$, and similarly for r_Z . Moreover $\tau^+(Z) - \tau^-(Z) \geq 1$.*

Proof. If a side of Z is parallel to the t -axis, then l_Z and r_Z are constant and $\tau^+(Z) - \tau^-(Z) = 1$. Otherwise let w be the vertex of Z with the least s . The two sides at w have directions $(\cos \gamma, \sin \gamma)$ and $(\sin \gamma, -\cos \gamma)$ for some $0 < \gamma < \frac{\pi}{2}$. The point $(l_Z(t), t)$ lies on the first side for $t \geq t(w)$ and on the second for $t \leq t(w)$, where ds/dt equals $\cot \gamma = \tan(\frac{\pi}{2} - \gamma)$ and $-\tan \gamma = \tan(-\gamma)$, respectively; the slope increases at $t(w)$ by $\cot \gamma + \tan \gamma = 1/(\sin \gamma \cos \gamma) \geq 2$, and l_Z is convex. The case of r_Z is symmetric, and $\tau^+(Z) - \tau^-(Z) = \cos \gamma + \sin \gamma \geq 1$. $\square$

For a square S_1 and S_2 either another square or S_0 , let L be the segment from the centre c_1 of S_1 to the centre c_2 of S_2 , respectively to a point c_2 of S_0 nearest to c_1 (any choice will do), and let $R = R(S_1, S_2)$ be the closed rectangle of points at distance at most $\frac{1}{2}$ from the line through L whose orthogonal projection lies on L . This is the rectangle used by Roth and Vaughan [25, §3, p. 172]; the following proposition replaces their constant $c = 10^{-10}$ by an explicit, much larger one. Let u be the unit vector along L and n a unit normal, and use the coordinates (s, t) of $c_1 + su + tn$; thus $R = \{0 \leq s \leq |L|, |t| \leq \frac{1}{2}\}$.

Proposition 4.3 (sharpened fundamental lemma). *Let $\theta := \theta(S_1, S_2)$, and let m' be the number of squares $Z \notin \{S_1, S_2\}$ of the packing that have a t -extreme vertex $v \in R$ with $|t(v)| < \frac{1}{2}$. Let $\mathcal{T} \subset (-\frac{1}{2}, \frac{1}{2})$ be the set of the numbers $t(v)$ for these vertices, and let $\lambda_1, \dots, \lambda_q$ be the lengths of the intervals into which the points of $\mathcal{T}$ cut $(-\frac{1}{2}, \frac{1}{2})$. Then the waste in R has area at least $\frac{\theta(2-\theta)}{4} \sum_r \lambda_r^2 \geq \theta(2-\theta)/(4(m'+1))$.*

Proof. Centres of squares lie in the convex set $[\frac{1}{2}, k - \frac{1}{2}]^2$. If S_2 is a square, L lies in this set and R lies within $\frac{1}{2}$ of it, so $R \subset [0, k]^2$. If $S_2 = S_0$, then L is perpendicular to a side of $[0, k]^2$ nearest to c_1 , and R is the rectangle of width 1 standing on that side, centred at the foot of L ; since $c_1 \in [\frac{1}{2}, k - \frac{1}{2}]^2$, again $R \subset [0, k]^2$, and $[0, k]^2$ lies in the half-plane $\{s \leq |L|\}$. Let $\Sigma := \{|t| < \frac{1}{2}\}$ be the open strip, λ_t the line at height t , and $\ell_t := \lambda_t \cap R$ for $|t| < \frac{1}{2}$. The points $c_1 + tn$ with $|t| < \frac{1}{2}$ are at distance less than $\frac{1}{2}$ from c_1 , hence interior points of S_1 ; likewise the points $c_2 + tn$ are interior points of S_2 if S_2 is a square.

(a) *Squares in the strip.* Let $Z \notin \{S_1, S_2\}$ be a square that meets some chord ℓ_{t_0} , $|t_0| < \frac{1}{2}$. The convex set $Z \cap \Sigma$ contains no interior point of S_1 or S_2 , so it misses the segment $\Sigma \cap \{s = 0\}$ and, if S_2 is a square, the segment $\Sigma \cap \{s = |L|\}$; it therefore lies on one side of each of them, and as it meets ℓ_{t_0} , we get $Z \cap \Sigma \subset R$ (in the wall case we also use $Z \subset \{s \leq |L|\}$). Hence for $|t| < \frac{1}{2}$ the square Z meets ℓ_t if and only if $\tau^-(Z) \leq t \leq \tau^+(Z)$, in the coordinates (s, t) . If $|\tau^+(Z)| < \frac{1}{2}$, a t -extreme vertex of Z at which $t = \tau^+(Z)$ lies in $Z \cap \Sigma \subset R$, so Z is one of the m' squares of the statement; the same holds for $\tau^-(Z)$. Since $\tau^+(Z) - \tau^-(Z) \geq 1$ (Lemma 4.2), at most one of $\tau^\pm(Z)$ lies in $(-\frac{1}{2}, \frac{1}{2})$.

Consequently the numbers $\tau^\pm(Z)$ in $(-\frac{1}{2}, \frac{1}{2})$, for all such Z , are exactly the points of $\mathcal{T}$, and $\mathcal{T}$ has at most m' elements. Cutting $(-\frac{1}{2}, \frac{1}{2})$ at the points of $\mathcal{T}$ gives $q \leq m' + 1$ open intervals I_r of lengths λ_r , with $\sum_r \lambda_r = 1$.

(b) *The same squares in the same order.* Fix r . By (a), for every square $Z \notin \{S_1, S_2\}$ either Z meets ℓ_t for all $t \in I_r$ or for none. Two disjoint compact convex sets that both meet all the lines λ_t , $t \in I_r$, are strictly separated by a line that is not parallel to u , so they appear in the same order on all these lines. Hence for $t \in I_r$ the chord ℓ_t meets the same squares in the same order, say $Z_0 = S_1, Z_1, \dots, Z_M$, followed by $Z_{M+1} = S_2$ (or the wall S_0). For $1 \leq i \leq M + 1$ and $t \in I_r$ let

$$g_i(t) := l_{Z_i}(t) - r_{Z_{i-1}}(t) \geq 0$$

(with $l_{Z_{M+1}} \equiv |L|$ in the wall case); this is the length of the uncovered part of ℓ_t between Z_{i-1} and Z_i , since by (a) the chord of each Z_i , $1 \leq i \leq M$, on the line λ_t lies in ℓ_t . The regions swept by these uncovered parts for different i and r are disjoint parts of the waste in R .

(c) *Each gap is convex.* By Lemma 4.2, g_i is convex and piecewise linear on I_r , and at each of its break points its slope increases by at least 2. Off the break points, $g'_i = \tan \psi' - \tan \psi$, where $\psi, \psi' \in (-\frac{\pi}{2}, \frac{\pi}{2})$ are the angles between n and the sides of Z_{i-1} and Z_i (or the wall) bounding the gap, so

$$|g'_i| \geq |\psi' - \psi| \geq \varphi_i := \theta(Z_{i-1}, Z_i),$$

because the direction of a side of a square is congruent modulo $\pi/2$ to the orientation of the square. As $\varphi_i \leq \frac{\pi}{4} < 1$, Lemma 4.1 gives $\int_{I_r} g_i \geq \varphi_i(2 - \varphi_i)\lambda_r^2/4$.

(d) *Summation.* By the triangle inequality for inclinations, $\sum_{i=1}^{M+1} \varphi_i \geq \theta$. Put $\varphi'_i := \min(\varphi_i, \theta)$; then $\sum_i \varphi'_i \geq \theta$ (either some $\varphi_i \geq \theta$, or $\varphi'_i = \varphi_i$ for all i), and since $x(2 - x)$ is increasing on $[0, 1]$,

$$\sum_i \varphi_i(2 - \varphi_i) \geq \sum_i \varphi'_i(2 - \varphi'_i) \geq (2 - \theta) \sum_i \varphi'_i \geq \theta(2 - \theta).$$

Summing over i and r , the waste in R has area at least

$$\sum_r \frac{\theta(2 - \theta)\lambda_r^2}{4} \geq \frac{\theta(2 - \theta)}{4q} \left(\sum_r \lambda_r \right)^2 \geq \frac{\theta(2 - \theta)}{4(m' + 1)}. \quad \square$$

For nearly parallel, nearly touching pairs the bound $\sum_r \lambda_r^2 \geq 1/(m' + 1)$ is far from the truth: the cut points can only come from squares wedged into the two small triangles left free by S_1 and S_2 in R , and few of them fit there. We first record an elementary fact about sections of a square.

Lemma 4.4 (sections near a vertex). *Let Z be a square, e a unit vector, and v a vertex of Z at which $p \mapsto p \cdot e$ is smallest on Z . For $0 \leq \nu \leq \frac{3}{4}$ the line $\{p : p \cdot e = v \cdot e + \nu\}$ meets Z in a segment of length at least $\min(2\nu, 1)$.*

Proof. Measure heights by $p \cdot e - v \cdot e$, and let $0 \leq \psi \leq \frac{\pi}{2}$ be the angle between one side at v and the line $\{p \cdot e = v \cdot e\}$; the neighbours of v are at heights $\sin \psi$ and $\cos \psi$, and the opposite vertex at $\sin \psi + \cos \psi \geq 1$. If $\psi \in \{0, \frac{\pi}{2}\}$ every section at height $\nu \in (0, 1]$ has length 1. Otherwise the section at height ν has length $\nu/(\sin \psi \cos \psi) \geq 2\nu$ for $\nu \leq \min(\sin \psi, \cos \psi)$, length $1/\max(\sin \psi, \cos \psi) \geq 1$ for ν between $\min(\sin \psi, \cos \psi)$ and $\max(\sin \psi, \cos \psi)$, and length $(\sin \psi + \cos \psi - \nu)/(\sin \psi \cos \psi)$ for larger ν . In the last case, if $\nu \leq \frac{3}{4}$, then $\sin \psi, \cos \psi \leq \frac{3}{4}$; with $p := \sin \psi + \cos \psi \in [1, \sqrt{2}]$ we get $p - \sin \psi \cos \psi = 1 - \frac{1}{2}(p - 1)^2 \geq 0.91 > \nu$, so the length exceeds 1. $\square$

Proposition 4.5 (nearly parallel pairs). *Let S_1 be a square and S_2 a square or S_0 , at distance less than d , with $\theta := \theta(S_1, S_2) \leq \theta_1$, where $d, \theta_1 \leq 10^{-5}$. Then $\sum_r \lambda_r^2 \geq Q_* := 0.32$ in Proposition 4.3, so the waste in $R(S_1, S_2)$ has area at least $Q_*\theta(2 - \theta)/4$.*

Proof. Put $\varepsilon_1 := d + \theta_1 \leq 2 \cdot 10^{-5}$ and $\varepsilon_0 := 10^{-4}$. We use the notation of the proof of Proposition 4.3; recall that every square Z contributing a point of $\mathcal{T}$ satisfies $Z \cap \Sigma \subset R$ and $Z \cap (S_1 \cup S_2) = \emptyset$. A point $\tau \in \mathcal{T}$ is *deep* if $|\tau| \leq \frac{1}{2} - \varepsilon_0$. For each deep τ fix a square Z with $\tau^-(Z) = \tau$ or $\tau^+(Z) = \tau$; we call τ an *upper* point in the first case, when Z lies in $\{t \geq \tau\}$, with *depth* $\frac{1}{2} - \tau$, and a *lower* point in the second case, with *depth* $\tau + \frac{1}{2}$. Different points come from different squares, since at most one of $\tau^\pm(Z)$ lies in $(-\frac{1}{2}, \frac{1}{2})$. For t in the t -range of a set X let $J_X(t)$ be the section of X by the line $\lambda_t = \{p : t(p) = t\}$.

(a) *Shallow points.* Let λ'_r be the lengths of the intervals into which the deep points cut $(-\frac{1}{2}, \frac{1}{2})$. Adding the shallow points (those in $(\frac{1}{2} - \varepsilon_0, \frac{1}{2})$ or $(-\frac{1}{2}, -\frac{1}{2} + \varepsilon_0)$) cuts only the top and the bottom interval, say of lengths $m_+ \leq 1$ and $m_- \leq 1$; parts of length at least $m_+ - \varepsilon_0$ and $m_- - \varepsilon_0$ remain uncut, so $\sum_r \lambda_r^2 \geq \sum_r \lambda_r'^2 - 4\varepsilon_0$ (if top and bottom interval coincide, the bound is $(1 - 2\varepsilon_0)^2$).

(b) *The wall case.* Let $S_2 = S_0$. For each side of $[0, k]^2$ the distance from S_1 to that side equals the distance from c_1 to it minus $\frac{1}{2}(\cos \theta + \sin \theta)$, so the side on which R stands is also a side nearest to S_1 . In coordinates in which $R = [-\frac{1}{2}, \frac{1}{2}] \times [0, |L|]$ stands on the x -axis and $t(p) = x$, S_1 has inclination θ and centre $(0, |L|)$, so it contains the square $(0, |L|) + [-\frac{\sigma_\theta}{2}, \frac{\sigma_\theta}{2}]^2$

with $\sigma_\theta := 1/(\cos \theta + \sin \theta) \geq 1 - \theta$; its lowest point is at height $< d$, so $|L| < d + \frac{1}{2}(\cos \theta + \sin \theta)$ and $R \setminus S_1 \subset \{y < |L| - \frac{\sigma_\theta}{2}\} \cup \{|x| > \frac{\sigma_\theta}{2}\}$, where $|L| - \frac{\sigma_\theta}{2} < d + \theta \leq \varepsilon_1$. If $\tau = \tau^-(Z)$ were deep, Lemma 4.4 would give, for $0 < \nu \leq \varepsilon_0/2$, a section of Z of length $\geq 2\nu$ on the line $x = \tau + \nu$, which satisfies $|x| \leq \frac{1}{2} - \frac{\varepsilon_0}{2} < \frac{1-\theta}{2} \leq \frac{\sigma_\theta}{2}$ (as $\theta < \varepsilon_0$); this section lies in $R \setminus S_1$, hence in $\{y < \varepsilon_1\}$, which is impossible for $\nu = \varepsilon_0/2 > \varepsilon_1/2$. The same holds for τ^+ . So there are no deep points, and $\sum \lambda_r^2 \geq (1 - 2\varepsilon_0)^2 > Q_*$.

(c) *Normal form.* Let now S_2 be a square. Two disjoint convex polygons are separated by a line parallel to a side of one of them (the origin lies outside the convex polygon $S_1 - S_2$, whose sides are parallel to sides of S_1 or S_2 , so it violates one of the inequalities defining that polygon). Exchanging S_1 and S_2 if necessary (this does not change R or $\sum \lambda_r^2$) and choosing coordinates, we may assume $S_1 = [-\frac{1}{2}, \frac{1}{2}]^2$, $S_2 \subset \{x \geq \frac{1}{2}\}$, and $c_2 = (a, b)$ with $b \geq 0$. Then S_2 has inclination θ , its projections to the axes have length $w := \cos \theta + \sin \theta \in [1, 1 + \theta]$, and S_2 contains the axis-parallel square $S_2^\circ := c_2 + [-\frac{\sigma_\theta}{2}, \frac{\sigma_\theta}{2}]^2$, $\sigma_\theta := 1/w \geq 1 - \theta$. Since $S_2 \subset \{x \geq \frac{1}{2}\}$ and some point of S_2 is within d of S_1 , we have $1 \leq a < 1 + d + \frac{\theta}{2}$, $0 \leq b < 1 + d + \frac{\theta}{2}$, and $\ell := a - \frac{\sigma_\theta}{2}$, the abscissa of the left side of S_2° , satisfies $\frac{1}{2} \leq \ell < \frac{1}{2} + d + \theta$. Put $h := b/a \in [0, 1 + \varepsilon_1]$ and $\rho_h := \sqrt{1 + h^2}$, so that $|L| = a\rho_h$, and take $n = (-b, a)/|L|$, so that $t(x, y) = (ay - bx)/|L|$. A direct computation gives the levels of the vertices $V_1 = (\frac{1}{2}, \frac{1}{2})$, $V_2 = (-\frac{1}{2}, \frac{1}{2})$, $V_3 = (\frac{1}{2}, -\frac{1}{2})$ of S_1 and $V'_1 = (\ell, b - \frac{\sigma_\theta}{2})$, $V'_2 = (\ell, b + \frac{\sigma_\theta}{2})$, $V'_3 = (\ell + \sigma_\theta, b - \frac{\sigma_\theta}{2})$ of S_2° :

$$\begin{aligned} t(V_1) &= \frac{1-h}{2\rho_h}, & t(V_2) &= \frac{1+h}{2\rho_h}, & t(V_3) &= -\frac{1+h}{2\rho_h}, \\ t(V'_1) &= -\sigma_\theta \frac{1-h}{2\rho_h}, & t(V'_2) &= \sigma_\theta \frac{1+h}{2\rho_h}, & t(V'_3) &= -\sigma_\theta \frac{1+h}{2\rho_h}. \end{aligned}$$

On the line λ_t , with s as coordinate, the lines $x = \frac{1}{2}$, $x = \ell$, $y = \frac{1}{2}$, $y = b - \frac{\sigma_\theta}{2}$ are met at $s = \frac{\rho_h}{2} + ht$, $s = \rho_h \ell + ht$, $s = (\frac{\rho_h}{2} - t)/h$ and $s = (\rho_h(b - \frac{\sigma_\theta}{2}) - t)/h$ respectively ($h > 0$).

(d) *The free gap.* For $|t| < \frac{1}{2}$ the points $c_1 + tn$ and $c_2 + tn$ lie in the interiors of S_1 and S_2 , and $J_R(t)$ is the segment between them. Hence, if Z contributes a point of $\mathcal{T}$ and $J_Z(t) \neq \emptyset$ with $|t| < \frac{1}{2}$, then $J_Z(t) \subset J_R(t) \setminus (S_1 \cup S_2)$ lies between $J_{S_1}(t)$ and $J_{S_2}(t) \supset J_{S_2^\circ}(t)$, and $|J_Z(t)| \leq \Gamma(t)$, where $\Gamma(t)$ denotes the distance from the right end of $J_{S_1}(t)$ to the left end of $J_{S_2^\circ}(t)$. The right end of $J_{S_1}(t)$ lies on $x = \frac{1}{2}$ for $t(V_3) \leq t \leq t(V_1)$ and on $y = \frac{1}{2}$ for $t(V_1) \leq t \leq t(V_2)$; the left end of $J_{S_2^\circ}(t)$ lies on $y = b - \frac{\sigma_\theta}{2}$ for $t(V'_3) \leq t \leq t(V'_1)$ and on $x = \ell$ for $t(V'_1) \leq t \leq t(V'_2)$. If $h \leq 1$, put $t_{lo} := t(V'_1)$ and $t_{hi} := \min(t(V_1), t(V'_2))$; for $t \in [t_{lo}, t_{hi}]$ the two ends lie on $x = \frac{1}{2}$ and $x = \ell$, so $\Gamma(t) = \rho_h(\ell - \frac{1}{2})$. If $h > 1$, put $t_{lo} := t(V_1) < 0$ and $t_{hi} := t(V'_1)$; for $t \in [t_{lo}, t_{hi}]$ they lie on $y = \frac{1}{2}$ and $y = b - \frac{\sigma_\theta}{2}$, so $\Gamma(t) = \rho_h(b - \frac{\sigma_\theta}{2} - \frac{1}{2})/h$. In both cases $\Gamma \leq \gamma_* := 1.415\varepsilon_1$ on $[t_{lo}, t_{hi}]$, since $\ell - \frac{1}{2} < \varepsilon_1$, $b - \frac{\sigma_\theta}{2} - \frac{1}{2} < \varepsilon_1$, $\rho_h \leq 1.415$, and $\rho_h/h \leq \sqrt{2}$ for $h > 1$.

(e) *Depths.* Let $\tau = \tau^-(Z)$ be an upper point and suppose $\tau < t_{hi} - 1.5\gamma_*$. Put $t' := \tau + 0.75\gamma_*$ if this is $\geq t_{lo}$, and $t' := t_{lo}$ otherwise. Then $t' \in [t_{lo}, t_{hi}]$, $|t'| < \frac{1}{2}$, and $0.75\gamma_* \leq t' - \tau < \frac{1}{2}$ (note $t_{lo} \leq 0$ and $\tau \geq -\frac{1}{2} + \varepsilon_0$), so by Lemma 4.4 and (d), $\gamma_* \geq \Gamma(t') \geq |J_Z(t')| \geq \min(2(t' - \tau), 1) \geq 1.5\gamma_*$, a contradiction. Hence every upper point has depth at most $D_u := \frac{1}{2} - t_{hi} + 1.5\gamma_*$, and symmetrically (using $t' := \tau - 0.75\gamma_*$ or t_{hi}) every lower point has depth at most $D_l := \frac{1}{2} + t_{lo} + 1.5\gamma_*$; in particular $D_u + D_l = 1 - (t_{hi} - t_{lo}) + 3\gamma_* \leq 1 + \beta_1$ with $\beta_1 := 3\gamma_*$. Let $D(h) := \frac{1}{2} - \frac{1-h}{2\rho_h}$; it is increasing, with derivative $(1+h)/(2\rho_h^3)$, and $D(h) \geq \frac{1}{2}$ for $h \geq 1$. By (c), if $h \leq 1$ then $\frac{1}{2} - t(V_1) = D(h)$, $\frac{1}{2} - t(V'_2) \leq \frac{1}{2} - \frac{1+h}{2\rho_h} + \frac{1+h}{2\rho_h}\theta \leq 0.71\theta$ (as $(1+h)/\rho_h \leq \sqrt{2}$ for $0 \leq h \leq 1$) and $\frac{1}{2} + t(V'_1) \leq D(h) + \frac{\theta}{2}$, while if $h > 1$ then $\frac{1}{2} - t(V'_1)$ and $\frac{1}{2} + t(V_1)$ are at most $\frac{1}{2} \leq D(h)$; so in all cases $D_u, D_l \leq D(h) + 0.71\theta + 1.5\gamma_* \leq D(h) + 3\varepsilon_1$ (as $\theta \leq \varepsilon_1$ and $1.5\gamma_* < 2.13\varepsilon_1$).

(f) *Two points on the same side.* Suppose $D(h) \geq 0.19$; then $h \geq 0.344$ (as $D(0.344) < 0.19$), so that $\tau_* := t(V'_2) - \frac{1}{2} > 0$ and $t(V_1) \leq t(V'_2) \leq t(V_2)$. Let Z give an upper point τ of depth ζ .

Then

$$t(V'_2) - \tau = \zeta + \tau_* \leq D_u + \tau_* \leq D(h) + \frac{1+h}{2\rho_h} - \frac{1}{2} + 3\varepsilon_1 = \frac{h}{\rho_h} + 3\varepsilon_1 < \frac{3}{4},$$

and $\tau \geq \frac{1}{2} - D_u > -0.02$. Hence for $t \in [\tau, t(V'_2)]$ each of Z , S_1 , S_2° has a nonempty section on λ_t (the t -ranges of S_1 and S_2° contain $[t(V'_3), t(V'_2)] \supset [-0.6, t(V'_2)]$, and that of Z contains $[\tau, \tau + 1]$); the end points of these sections depend continuously on t and the three sets are pairwise disjoint, so their order on λ_t does not depend on t . By (d), applied at a level in $[\tau, \frac{1}{2})$, $J_Z(t)$ lies between $J_{S_1}(t)$ and $J_{S_2^\circ}(t)$ for all $t \in [\tau, t(V'_2)]$. At the level $t(V'_2)$ the section of S_2° is the point V'_2 , and that of S_1 ends on the side $y = \frac{1}{2}$; by (c) the gap between them is

$$\Gamma_* := \rho_h \left(\ell - \frac{1}{2} \right) + \frac{\rho_h}{2h} (h - 1 + \sigma_\theta(1 + h)) \leq \rho_h + \rho_h \left(\ell - \frac{1}{2} \right) < 2,$$

and by Lemma 4.4 the section $J_Z(t(V'_2))$ has length at least $\min(2(\zeta + \tau_*), 1)$. Now let two upper points of depths $\zeta' \leq \zeta$ come from squares Z', Z . Their sections at the level $t(V'_2)$ are disjoint and lie in a segment of length Γ_* ; hence $\zeta' + \tau_* < \frac{1}{2}$. If $\zeta + \tau_* \leq \frac{1}{2}$, then $2(\zeta' + \tau_*) + 2(\zeta + \tau_*) \leq \Gamma_*$, that is, $\zeta' \leq \chi_u - \zeta$ with $\chi_u := \frac{1}{2}\Gamma_* - 2\tau_*$. If $\zeta + \tau_* > \frac{1}{2}$, then $2(\zeta' + \tau_*) + 1 \leq \Gamma_*$, so

$$\zeta' \leq \frac{\Gamma_*}{2} - t(V'_2) = \frac{h(h-1)}{2\rho_h} + \frac{\rho_h}{2} \left(\ell - \frac{1}{2} \right) - \frac{(1-\sigma_\theta)(1+h)(1-h)^2}{4h\rho_h} < 1.1\varepsilon_1 < \varepsilon_0,$$

which is impossible for a deep point; so in this case there is at most one upper point. With $\chi(h) := 1 + \frac{\rho_h}{2} - \frac{1+h}{\rho_h}$ we get $\chi_u \leq \chi(h) + (1-\sigma_\theta)\frac{1+h}{\rho_h} + \frac{\rho_h}{2}(\ell - \frac{1}{2}) \leq \chi(h) + 2.2\varepsilon_1$. The lower points are treated in the same way below the strip, down to the level $-\frac{1}{2} - \tau_* = t(V'_3)$, where the section of S_2° is the point V'_3 and that of S_1 ends on $x = \frac{1}{2}$ (as $t(V_3) \leq t(V'_3) \leq t(V_1)$), at distance $\Gamma_l := \rho_h(a - \frac{1-\sigma_\theta}{2}) \leq \rho_h + \rho_h(a-1)$. For two lower points of depths $\zeta' \leq \zeta$ this gives $\zeta' \leq \chi_l - \zeta$ if $\zeta + \tau_* \leq \frac{1}{2}$, where $\chi_l := \frac{1}{2}\Gamma_l - 2\tau_* \leq \chi(h) + (1-\sigma_\theta)\frac{1+h}{\rho_h} + \frac{\rho_h}{2}(a-1) \leq \chi(h) + 2.2\varepsilon_1$, and $\zeta' \leq \frac{1}{2}\Gamma_l - \frac{1}{2} - \tau_* \leq \frac{h(h-1)}{2\rho_h} + \frac{\rho_h}{2}(a-1) + (1-\sigma_\theta)\frac{1+h}{2\rho_h} < 1.8\varepsilon_1 < \varepsilon_0$ otherwise.

(g) *Conclusion.* Let $\zeta_u \geq \zeta'_u$ ($\zeta_l \geq \zeta'_l$) be the largest and the second largest depth of an upper (lower) point, with the value 0 if there is no such point. Consider the intervals $(\frac{1}{2} - \zeta_u + \beta_1, \frac{1}{2} - \zeta'_u)$, $(-\frac{1}{2} + \zeta'_l, -\frac{1}{2} + \zeta_l - \beta_1)$ and $(-\frac{1}{2} + \zeta_l, \frac{1}{2} - \zeta_u)$ (some may be empty). They contain no deep point: upper points have depth ζ_u , ζ'_u or less than ζ'_u , and lower points lie at levels $\leq -\frac{1}{2} + D_l \leq \frac{1}{2} - D_u + \beta_1 \leq \frac{1}{2} - \zeta_u + \beta_1$; symmetrically for the second interval. They are pairwise disjoint, by the same inequality. Hence, with $\mu_u := \zeta_u - \zeta'_u$ and $\mu_l := \zeta_l - \zeta'_l$,

$$\sum_r \lambda_r^2 \geq (1 - \zeta_u - \zeta_l)_+^2 + (\mu_u - \beta_1)_+^2 + (\mu_l - \beta_1)_+^2 \geq (1 - \zeta_u - \zeta_l)_+^2 + \mu_u^2 + \mu_l^2 - 2.04\beta_1,$$

since $(\mu - \beta_1)_+^2 \geq \mu^2 - 2\beta_1\mu$ and $\mu \leq D_u, D_l \leq 0.51$. If $D(h) < 0.19$, then $\zeta_u, \zeta_l < 0.1906$ and $\sum \lambda_r^2 \geq (1 - 0.3812)^2 > 0.38$, so by (a) $\sum_r \lambda_r^2 \geq 0.38 - 4\varepsilon_0 > Q_*$. Let $D(h) \geq 0.19$ and $\chi_* := \max(\chi_u, \chi_l)$. By (f), on each side either $\mu = \zeta$ (at most one point) or $\mu \geq (2\zeta - \chi_*)_+$; in both cases $\mu^2 \geq \xi(\zeta) := \min\{\zeta^2, ((2\zeta - \chi_*)_+)^2\}$. Hence $\sum \lambda_r^2 \geq \Xi(\zeta_u, \zeta_l) - 2.04\beta_1$ with $\Xi(z, z') := (1 - z - z')_+^2 + \xi(z) + \xi(z')$. The function χ decreases on $[0, h_c]$ and increases on $[h_c, \infty)$, where $h_c^3 + 3h_c - 2 = 0$ (indeed $\chi'(h) = (h^3 + 3h - 2)/(2\rho_h^3)$), so $\chi_* \leq \max(\chi(0.344), \chi(1.00002)) + 2.2\varepsilon_1 < 0.29295$. If $z + z' > 1$, then $z, z' > 0.49 > \chi_*$ (as $z, z' \leq \max(D_u, D_l) \leq D(h) + 3\varepsilon_1 \leq 0.51$ by (e)) and $\Xi(z, z') \geq z^2 + z'^2 > 0.48$. Otherwise, choosing one of the two expressions for each of $\xi(z)$ and $\xi(z')$, Ξ is bounded below by the least of: $(1 - z - z')^2 + z^2 + z'^2 \geq \frac{1}{3}$; $(1 - z - z')^2 + z^2 + ((2z' - \chi_*)_+)^2 \geq \frac{1}{2}(1 - z')^2 + ((2z' - \chi_*)_+)^2 \geq \frac{1}{9}(2 - \chi_*)^2$ (the minimum is at $z' = \frac{1+4\chi_*}{9}$; for $z' \leq \frac{\chi_*}{2}$ the bound is $\frac{1}{2}(1 - \frac{\chi_*}{2})^2$, which is larger); and $(1 - z - z')^2 + ((2z - \chi_*)_+)^2 + ((2z' - \chi_*)_+)^2 \geq \frac{2}{3}(1 - \chi_*)^2$ (a convex symmetric function, minimal at $z = z' = \frac{1+2\chi_*}{6}$). For $\chi_* < 0.29295$ all

these exceed 0.3237, so $\sum \lambda_r^2 \geq 0.3237 - 2.04\beta_1 > 0.3235$ (as $\beta_1 = 3\gamma_* \leq 8.5 \cdot 10^{-5}$). By (a), $\sum_r \lambda_r^2 \geq 0.3235 - 4\epsilon_0 > Q_*$. $\square$

Remark 4.6. Roth and Vaughan [25, §3, pp. 172–173] assume that S_1 and S_2 are at distance at most 1, bound the number m of squares meeting R by 20, and use a single strip of width $1/(4m+1)$ parallel to L that contains no vertex in its interior, and a single trapezoid in it; this gives the lower bound $(4m+1)^{-2} \tan(\theta/2m)$, about $3.8 \cdot 10^{-6} \theta$ for $m = 20$, and their lemma is stated with $c = 10^{-10}$ [25, p. 171, (3)]. In Proposition 4.3 the strip is cut only where the set of squares met by the chords changes, the gaps are allowed to be convex rather than affine, and all sub-bands and gaps are summed; with the count of Lemma 4.8(a) this gives $\theta(2-\theta)/40 \geq \theta/33$ for pairs at distance less than 10^{-4} , and Proposition 4.5 gives $0.32\theta(2-\theta)/4 \geq \theta/6.26$ for pairs at distance less than 10^{-5} that are moreover nearly parallel. In randomized adversarial searches (with S_2 perturbed as in Proposition 4.5) we found no configuration with $\sum \lambda_r^2 < \frac{1}{3}$; the value $\frac{1}{3}$ is attained by two squares wedged into the two triangles at depth $\frac{1}{3}$, and two deep points never seemed to share a triangle. Roth and Vaughan’s standing assumption $\alpha(\alpha - [\alpha]) > 1/6$ on the side of the container plays no role here. A version of their lemma for squares joined by a path, with a constant derived from theirs, is given in [5, Section 4].

Let $\mathcal{V}_h$ consist of the squares of the packing together with the left and the right side of $[0, k]^2$. Two distinct members P, Q of $\mathcal{V}_h$ are *consecutive on the line y* ($0 < y < k$) if both meet the segment $[0, k] \times \{y\}$ and the open segment between them on that line meets no square. Let G_h be the graph on $\mathcal{V}_h$ in which P and Q are adjacent if they are consecutive on a set of lines y of positive measure. Define $\mathcal{V}_v$ and G_v in the same way, with vertical lines and the bottom and the top side of $[0, k]^2$.

Lemma 4.7 (visibility graphs are planar). *The graphs G_h and G_v are planar.*

Proof. We treat G_h . For every edge $e = \{P, Q\}$ choose a line $y_e \in (0, k)$ on which P and Q are consecutive, with $y_e \neq y_{e'}$ for $e \neq e'$ (each edge has a set of such lines of positive measure). Let P be to the left of Q on y_e , and let $a_e \in P$ and $b_e \in Q$ be the end points of the segment between them; $a_e \neq b_e$ if both are squares, since they are disjoint (if one of them is a side, possibly $a_e = b_e$, and the middle piece of the path below is empty). Represent a square by its centre, the left side by $o_- := (-2, k/2)$ and the right side by $o_+ := (k+2, k/2)$, and draw e as the path from the representative of P to a_e , along $[a_e, b_e]$, and on to the representative of Q . Here the piece inside a square is the segment from its centre to the boundary point; it lies in the square, and all its points except the last one are interior points. The piece for the left side is the segment from o_- to $(-1, y_e)$ followed by the segment from $(-1, y_e)$ to $(0, y_e) = a_e$, and symmetrically for the right side. Two such paths meet only in common end points. Indeed, segments from the centre of a square to distinct boundary points (the points a_e, b_e of different edges lie on different lines) meet only at the centre, and different squares are disjoint. The open segments (a_e, b_e) lie on distinct lines and meet no square. The pieces for the sides lie outside $(0, k) \times \mathbb{R}$, the horizontal ones lie on distinct lines, and the segments from o_- to the distinct points $(-1, y_e)$ meet only at o_- (likewise at o_+). So this is a plane drawing of G_h . $\square$

For $0 < d \leq 10^{-4}$ put $\Lambda = \Lambda(d) := d + \sqrt{2}$ and $R_0 = R_0(d) := \sqrt{\Lambda^2 + \frac{1}{4}} < 1.5001$. We use the following theorem of Fodor [13, Theorem 1]: *the smallest circle that contains 12 points with mutual distances at least 1 has radius $R_{12} = 1.5148\dots$* (explicitly, $R_{12} = 1/(\sqrt{3}x_0)$, where x_0 is the smallest positive root of $9x^5 - 15x^4 + 7x^3 - 3x + 1$). We also use Oler’s inequality [24, Theorem 1] in the following Euclidean form, which is inequality (1) of Folkman and Graham [14], where it also receives a short proof: *a compact convex set $X \subset \mathbb{R}^2$ contains at most $\frac{2}{\sqrt{3}}A(X) + \frac{1}{2}P(X) + 1$ points with mutual distances at least 1, where A and P denote area and*

perimeter. (Oler's theorem is stated for the Minkowski distance of a convex, centrally symmetric Jordan curve, with the area divided by the critical determinant, which is $\sqrt{3}/2$ for the Euclidean norm, and for finite sets that lie in the closed region bounded by a Jordan polygon all of whose vertices belong to the set. The Euclidean form above follows by applying it to the boundary of the convex hull of the points, whose area and perimeter do not exceed those of X ; collinear points are trivial. Folkman and Graham [14, p. 751] make the same convex-hull reduction from their own inequality for planar simplicial complexes, which gives an independent proof of the form above.) Put

$$K := 27, \quad q_0 := 10.$$

Lemma 4.8 (counting). *Let $0 < d \leq 10^{-4}$.*

- (a) *If S_1 is a square and S_2 is a square or S_0 , at distance $< d$, then $m' \leq q_0 - 1 = 9$ in Proposition 4.3; hence the waste in $R(S_1, S_2)$ has area at least $\theta(2 - \theta)/(4q_0)$, $\theta = \theta(S_1, S_2)$.*
- (b) *Let $k \geq 4$ and let G be G_h or G_v . For an edge $e = \{P, Q\}$ of G whose members are at distance $< d$ put $R_e := R(P, Q)$ if both are squares, and $R_e := R(P, S_0)$ if P is a square and Q a side. Then every point lies in R_e for at most K such edges e (the two sides of $[0, k]^2$ that are vertices of G are at distance $k > d$).*

Proof. Every point of a unit square is within $\frac{\sqrt{2}}{2}$ of its centre, so the centres of two squares at distance $< d$ are at distance $< \Lambda$, and the centre of a square at distance $< d$ from S_0 is at distance $< d + \frac{\sqrt{2}}{2}$ from S_0 .

(a) For each of the m' squares Z of Proposition 4.3 choose a vertex v_Z as there, and let $Q_Z := Z \cap B(v_Z, 1)$; in coordinates in which $Z = [0, 1]^2$ and $v_Z = 0$ this is the quarter-disc $\{x, y \geq 0, x^2 + y^2 \leq 1\}$, of area $\frac{\pi}{4}$. The sets Q_Z , S_1 and (if it is a square) S_2 are pairwise disjoint, and they lie in the closed 1-neighbourhood $R + B(1)$ of R , because $v_Z, c_1, c_2 \in R$ and $S_i \subset B(c_i, \frac{\sqrt{2}}{2})$. By Steiner's formula, $R + B(1)$ has area $|L| + 2(|L| + 1) + \pi$. If S_2 is a square, then $|L| < \Lambda$ and $\frac{\pi}{4}m' + 2 \leq 3|L| + 2 + \pi$, so $m' \leq \frac{12}{\pi}|L| + 4 < \frac{12}{\pi}\Lambda + 4 < 10$. If $S_2 = S_0$, then $|L| < d + \frac{\sqrt{2}}{2}$, and all these sets lie in $[0, k]^2$, hence in the closed half-plane $\mathcal{H}$ that is bounded by the line through the side of $[0, k]^2$ on which R stands and contains $[0, k]^2$. The part of $R + B(1)$ in $\mathcal{H}$ has area $3|L| + 1 + \frac{\pi}{2}$ (we remove a 1×1 rectangle and two quarter-discs of radius 1 beyond that line), whence $\frac{\pi}{4}m' + 1 \leq 3|L| + 1 + \frac{\pi}{2}$ and $m' \leq \frac{12}{\pi}|L| + 2 < 5$. Now apply Proposition 4.3.

(b) If $p \in R_e$, the projection of p on the line of L lies on L , so p is within $\sqrt{|L|^2 + \frac{1}{4}}$ of both end points of L . For an edge joining two squares, both centres therefore lie in the open disc $B(p, R_0)$. For an edge joining a square P and a side Q , the square is at distance $< d$ from $S_0 \supset Q$, so (a) applies to (P, S_0) , and the centre of P lies within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}} < R_0$ of p . Disjoint squares contain the disjoint open discs of radius $\frac{1}{2}$ about their centres, so the centres are at mutual distance at least 1; as $R_0 < R_{12}$, at most 11 squares have their centre in $B(p, R_0)$. Recall that a simple planar graph with v vertices has at most $\max(1, 3v - 6)$ edges, and that the edges in question are edges of the planar graph G (Lemma 4.7). If none of them contains a side of $[0, k]^2$, they span at most 11 vertices, so there are at most $3 \cdot 11 - 6 = 27 = K$ of them.

Otherwise let $e_0 = \{P_0, Q_0\}$ be one of them with Q_0 a side, and let σ_0 be the side of $[0, k]^2$ on which $R_{e_0} = R(P_0, S_0)$ stands. Then p is at distance at most $|L_{e_0}| < d + \frac{\sqrt{2}}{2}$ from the line through σ_0 , while every centre is at distance at least $\frac{1}{2}$ from that line, on the same side as $[0, k]^2$. Hence, if ν is the unit normal of σ_0 pointing away from $[0, k]^2$, every centre z in question satisfies $(z - p) \cdot \nu < c_0 := d + \frac{\sqrt{2}}{2} - \frac{1}{2} < 0.20721$, so all these centres lie in the compact convex set $D_0 := \{z \in \overline{B}(p, R_0) : (z - p) \cdot \nu \leq c_0\}$. With $\phi := \arccos(c_0/R_0)$, this set has area

$R_0^2(\pi - \phi) + c_0\sqrt{R_0^2 - c_0^2} < 4.155$ and perimeter $2R_0(\pi - \phi) + 2\sqrt{R_0^2 - c_0^2} < 8.1$, so by Oler's inequality it contains at most $\frac{2}{\sqrt{3}} \cdot 4.155 + \frac{8.1}{2} + 1 < 9.85$, that is, at most 9, centres. Moreover, only one side of $[0, k]^2$ occurs among the edges in question: if $\{P, Q\}$ is one of them with Q the left side, then the centre of P is within $d + \frac{\sqrt{2}}{2}$ of the left side and p is within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}}$ of that centre, so p is within 1.58 of the left side; likewise for the right side (or for the bottom and the top side if $G = G_v$), and $k \geq 4 > 3.16$. So the edges span at most $9 + 1$ vertices, and there are at most $3 \cdot 10 - 6 = 24 < K$ of them. $\square$

Only the uncovered points of the rectangles R_e will matter below, and there the multiplicity is much smaller.

Lemma 4.9 (multiplicity at uncovered points). *Let $0 < d \leq 10^{-4}$, let G be G_h or G_v , and define R_e for the edges e of G whose members are at distance $< d$ as in Lemma 4.8(b). Then every point of the waste lies in R_e for at most $K_w := 13$ such edges.*

Proof. Let p be a point of the waste, $\Lambda := d + \sqrt{2}$, and call an edge e of G *relevant* if its members are at distance $< d$ and $p \in R_e$; let L_e be the segment defining R_e . Every square contains the closed disc of radius $\frac{1}{2}$ about its centre, so every centre is at distance $> \frac{1}{2}$ from p . As in the proof of Lemma 4.8(b), for a relevant e the orthogonal projection of p on the line of L_e lies on L_e , and p is at distance $\leq \frac{1}{2}$ from that line. If both members of e are squares, then $|L_e| < \Lambda$ and p is within $\sqrt{(\Lambda/2)^2 + \frac{1}{4}}$ of the nearer end point of L_e ; if e joins a square P and a side, then $|L_e| < d + \frac{\sqrt{2}}{2}$ and p is within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}}$ of the centre of P . As $\Lambda/2 \leq d + \frac{\sqrt{2}}{2}$, in both cases some member of e is a square whose centre lies within $\rho_1 := \sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}} < 0.8662$ of p ; we fix one such member and call its centre the *inner centre* of e . Put $f(r_1, r_2) := (r_1^2 + r_2^2 - 1)/(2r_1r_2)$; if two points are at distances r_1, r_2 from a point z and at distance ≥ 1 from each other, the cosine of the angle they subtend at z is at most $f(r_1, r_2)$.

(a) *Degree at a centre.* Let c be the centre of a square P , $r := |p - c|$, and let $e_1, \dots, e_m$ be the relevant edges containing P . Let c_j be the end point of L_{e_j} other than c : the centre of the other member, or, if the other member is a side, the point of S_0 used to define $R_{e_j} = R(P, S_0)$. The angle ψ_j between $p - c$ and $c_j - c$ satisfies $\cos \psi_j \geq 0$ and $r \sin \psi_j \leq \frac{1}{2}$, so $\psi_j \leq \arcsin \frac{1}{2r}$, and all directions $c_j - c$ lie in an arc of length $2 \arcsin \frac{1}{2r}$. At most one e_j joins P to a side: the two sides of $[0, k]^2$ that are vertices of G are at distance k , while P has width at most $\sqrt{2} < k - 2d$ in the perpendicular direction. If c_i and c_j are centres, they are at distances in $[1, \Lambda)$ from c (squares at distance $< d$) and at distance ≥ 1 from each other. For fixed $r_2 \geq 1$ the function $r_1 \mapsto f(r_1, r_2) = \frac{r_1}{2r_2} + \frac{r_2^2 - 1}{2r_2} \cdot \frac{1}{r_1}$ is convex, and symmetrically in r_2 , so the maximum of f on $[1, \Lambda]^2$ is attained at a corner, where f takes the values $\frac{1}{2}$, $\frac{\Lambda}{2}$ and $1 - \frac{1}{2\Lambda^2}$; hence the directions of $c_i - c$ and $c_j - c$ differ by at least $\gamma_0 := \arccos(1 - \frac{1}{2\Lambda^2})$. If c_i lies on a side σ of $[0, k]^2$, then $c_i - c$ is a positive multiple of the unit normal ν of σ pointing away from $[0, k]^2$, c is at distance less than $d + \frac{\sqrt{2}}{2}$ from the line of σ , and every centre c_j is at distance at least $\frac{1}{2}$ from it, on the same side; so $(c_j - c) \cdot \nu < d + \frac{\sqrt{2}}{2} - \frac{1}{2}$, and as $|c_j - c| \geq 1$ the two directions differ by more than $\arccos(d + \frac{\sqrt{2}}{2} - \frac{1}{2}) > 78^\circ > \gamma_0$. Consequently

$$m \leq D(r) := 1 + \left\lfloor \frac{2 \arcsin(1/(2r))}{\gamma_0} \right\rfloor,$$

and $D(r) \leq 5$ for $r > \frac{1}{2}$, since $\pi < 5\gamma_0$.

(b) *Counting.* Let $c_1, \dots, c_n$ be the distinct inner centres, at distances $r_i \in (\frac{1}{2}, \rho_1]$ from p . Every relevant edge contains the square of its inner centre, so by (a) there are at most $\sum_i D(r_i)$ relevant edges. Put $m_i := \max(2, D(r_i))$, and $R_2 := \rho_1$, $R_m := \min(\rho_1, 1/(2 \sin(\frac{(m-1)\gamma_0}{2}))$ for $m = 3, 4, 5$; then $r_i \leq R_{m_i}$. For $0 < r_1, r_2 < 1$ we have $\partial f / \partial r_1 = (r_1^2 - r_2^2 + 1)/(2r_1^2 r_2) > 0$, and symmetrically, so two inner centres c_i, c_j (at distance ≥ 1) are seen from p at an angle at least $\gamma(m_i, m_j)$, where $\gamma(a, b) := \arccos f(R_a, R_b)$. Order the inner centres by their direction from p . If $n \geq 3$, the angles between consecutive directions add up to 2π , so $\sum_i \gamma(m_i, m_{i+1}) \leq 2\pi$ (indices modulo n); if $n \leq 2$, then $\sum_i m_i \leq 10$. Since R_e does not depend on d , every edge that is relevant for d is relevant for 10^{-4} , so we may take $d = 10^{-4}$, where $\gamma_0 > 41.406^\circ$, $\rho_1 < 0.86611$, $R_3 < 0.75598$, $R_4 < 0.56571$, $R_5 < 0.50396$, and $\gamma(a, b) \geq \gamma(2, 2) > 70.52^\circ$, so $n \leq 5$. A finite check of all sequences $(m_1, \dots, m_n) \in \{2, 3, 4, 5\}^n$, $n \leq 5$, shows that every sequence with $\sum_i m_i \geq 15$ violates these angle conditions, with a margin of more than 25° , and that the only sequences with $\sum_i m_i = 14$ that satisfy them are, up to rotation, $(5, 2, 5, 2)$ (which uses about 359°). The check, in 50-digit arithmetic, is part of the accompanying script. So there are at most 14 relevant edges, and at most 13 unless two inner centres have class 5.

(c) *Two centres of class 5.* For an inner centre c write $\deg(c)$ for the number of relevant edges containing its square; the number of relevant edges is at most $\sum_i \deg(c_i)$, and an edge both of whose members are squares with centres among $c_1, \dots, c_n$ is counted twice in this sum. Suppose that two inner centres, say c_1 and c_3 , have class 5, i.e. $r_1, r_3 \leq R_5 < 0.50396$. Then $1 \leq |c_1 - c_3| \leq 2R_5 < 1.008$ and the angle at p between $c_1 - p$ and $c_3 - p$ is at least $\gamma(5, 5) > 165.6^\circ$, so the angles of the triangle pc_1c_3 at c_1 and at c_3 are acute, and by the law of sines the angle at c_1 between $p - c_1$ and $c_3 - c_1$ is at most $\psi := \arcsin(R_5 \sin(\pi - \gamma(5, 5))) < 7.19^\circ$. If c_3 is the other end point of a relevant edge at c_1 , this edge is counted in $\deg(c_1)$ and in $\deg(c_3)$, so there are at most $\sum_i D(r_i) - 1$ relevant edges. Otherwise let c_j be the other end point of a relevant edge at c_1 . If c_j is a centre, it is the centre of a square other than that of c_3 , so $|c_j - c_3| \geq 1$; as $|c_j - c_1|$ and $|c_3 - c_1|$ lie in $[1, \Lambda)$ (note $2R_5 < \Lambda$), the argument of (a) shows that the directions of $c_j - c_1$ and $c_3 - c_1$ differ by at least γ_0 . If c_j lies on a side, they differ by more than $78^\circ > \gamma_0$, as in (a). Hence the directions of the $c_j - c_1$ lie in the arc of half-width $\arcsin \frac{1}{2r_1} < 90^\circ$ about the direction of $p - c_1$ from which the open arc of half-width γ_0 about the direction of $c_3 - c_1$ has been removed, that is, in two closed arcs, each of length less than $90^\circ + 7.19^\circ - 41.406^\circ < 2\gamma_0$. Each of them contains at most two of these directions, so $\deg(c_1) \leq 4 = D(r_1) - 1$, and again there are at most $\sum_i D(r_i) - 1$ relevant edges. In all cases, by (b), there are at most $\sum_i D(r_i) - 1 \leq \sum_i m_i - 1 \leq 13$ relevant edges if two inner centres have class 5, and at most 13 otherwise. $\square$

The proof of Lemma 4.9 uses no property of G except that the only sides involved are two opposite ones; the bound holds for any family of such pairs at distance $< d$. A computer-assisted refinement of the same counting gives the bound 9, which is sharp for the number of all pairs at distance $< d$ (see the remark after the proof), though not known to be sharp for the edges of a single graph G .

Lemma 4.10 (multiplicity at uncovered points, computer-assisted). *Under the assumptions of Lemma 4.9, every point of the waste lies in R_e for at most $K_w^* := 9$ of the edges e considered there.*

Proof. As in the proof of Lemma 4.9 we may take $d = 10^{-4}$ and $p = 0$, and we keep the notation $\Lambda, \rho_1, \gamma_0$. Let I be the set of all centres within distance ρ_1 of p ; they satisfy $\frac{1}{2} < |c| \leq \rho_1$ and are pairwise at distance at least 1, so $n := |I| \leq 5$. By the proof of Lemma 4.9, every relevant edge contains a square with centre in I . At most E_{II} relevant edges join two squares with centres in I , where E_{II} is the number of pairs in I at distance less than Λ . Every other relevant edge joins a square with centre $c_i \in I$ either to a side of $[0, k]^2$ or to a square whose centre x satisfies

$|x| > \rho_1$, $1 \leq |x - c_i| < \Lambda$, $p \in R(c_i, x)$ and $|x - c_j| \geq 1$ for all $c_j \in I \setminus \{c_i\}$, where $R(c_i, x)$ denotes the rectangle R of the two squares with these centres; as in part (a) of the proof of Lemma 4.9, the directions of the vectors $x - c_i$ of these centres are pairwise at least γ_0 apart. Let $D_i^*(I)$ be the largest number of directions, pairwise at least γ_0 apart, in which such an x exists, and let W_I be the number of $c_i \in I$ that can belong to a relevant edge with a side (at most one per centre, since $\sqrt{2} < k - 2d$). Then the number of relevant edges is at most

$$U(I) := E_{II} + W_I + \sum_{c_i \in I} D_i^*(I).$$

A branch-and-bound computation over the parameters of I shows that a computed upper bound of $U(I)$ is at most 9 on every box of a cover of the parameter domain. Without a side, the centres are written in polar coordinates (r_i, ϑ_i) with $\vartheta_1 = 0 \leq \vartheta_2 \leq \dots \leq \vartheta_n$ (after a rotation and relabelling). If a relevant edge contains a side, its rectangle $R(P, S_0)$ stands on a side of $[0, k]^2$ whose line is at distance $h < d + \frac{\sqrt{2}}{2}$ from p , and the rectangles of all relevant edges with a side stand on one and the same side σ : two of them on opposite sides would force $k < 2d + \sqrt{2}$, which is impossible since $k \geq 4$, and if $R(P, \sigma)$ and $R(P', \sigma')$ with adjacent sides σ, σ' both contained p , then $P \neq P'$ (the rectangle of a square and a side is determined by the square) and, with the corner at the origin, the conditions $|p_x - c_x| \leq \frac{1}{2}$, $0 \leq p_x \leq c'_x < d + \frac{\sqrt{2}}{2}$, $c_x \geq \frac{1}{2}$ and the symmetric ones would give $|c - c'|^2 \leq \frac{1}{2}$, which is impossible. We take σ to be the line $y = -h$ and ignore the other sides (this only drops constraints); then every centre satisfies $y \geq -h + \frac{1}{2}$, no normalisation of ϑ_1 is used, and h is a further parameter. For a box of parameters the directions are divided into 1440 arcs of 0.25° and $|x - c_i|$ into 6 intervals; an arc is discarded only if for each interval one of the conditions $|x| > \rho_1$, $p \in R(c_i, x)$, $|x - c_j| \geq 1$ (and $y \geq -h + \frac{1}{2}$) fails for all parameters in the box and all points of the arc and the interval. Then $D_i^*(I)$ is bounded by the largest number of points, pairwise at least γ_0 apart, in the union of the remaining arcs (the largest of the circular greedy counts started at the beginning of each maximal run of arcs, or $\lfloor 2\pi/\gamma_0 \rfloor$ if no arc is discarded; this gives the maximum, since an optimal set can be rotated backwards, keeping all gaps, until one of its points reaches the start of its run, after which the j -th greedy point is never later than the j -th point of the rotated optimal set); E_{II} by the number of pairs that may be at distance less than Λ ; and W_I by the number of centres that may lie within $d + \frac{\sqrt{2}}{2}$ of the side with $|c_{i,x}| \leq \frac{1}{2}$ and $c_{i,y} \geq 0$. A box is accepted if it contains no admissible configuration (for every parameter of the box two centres are at distance less than 1, or a centre is closer than $\frac{1}{2}$ to the side), or, in the case with a side, if no side edge is possible (then it is covered by the case without a side), or if the bound is at most 9; otherwise it is bisected. For every $n = 1, \dots, 5$, with and without a side, the computation terminated with all boxes accepted (30,365 accepted boxes without a side and 48,308 with a side).

The search itself is not relied upon: it used floating-point interval bounds with a safety margin of 10^{-9} , and every accepted box was re-verified in ball (midpoint-radius interval) arithmetic with rigorous error bounds, using Arb [16], which has been part of FLINT [12] since version 3.0, through `python-flint` [17] (version 0.9.0, with FLINT 3.6.0, at 96-bit working precision), boxes being bisected further where the ball enclosures were too coarse (all 78,673 boxes confirmed, with 59,238 further bisections and largest rigorous bound 9). In this re-verification the arcs of directions are closed and cover the whole circle (the floating-point end point of each arc is replaced, where necessary, by the starting point of the next one, and the floating-point value $\widetilde{2\pi} < 2\pi$ by the exact 2π), and every interval of an angle ϑ_i ending at $\widetilde{2\pi}$ is extended to the exact 2π . This covers every admissible configuration with an angle in the arc $(\widetilde{2\pi}, 2\pi)$, which has length less than $2.5 \cdot 10^{-16}$: two centres with angles in this arc, or (without a side, where $\vartheta_1 = 0$) one such centre and c_1 , would be at distance less than $\rho_1 - \frac{1}{2} + 10^{-15} < 1$, so only ϑ_n can lie

in the arc, and only in the case with a side; replacing ϑ_n by $\widetilde{2\pi}$ then gives a configuration with ordered angles in the domain of the search, hence in an accepted box (the accepted boxes leave no gap, see below), whose ϑ_n -interval ends at $\widetilde{2\pi}$ and is therefore extended, and the extended box contains the original configuration. The final count of points pairwise at least γ_0 apart in a union of arcs is a finite combinatorial computation on the arc end points (the greedy count described above); it is done in double precision, with γ_0 decreased by 10^{-12} and the condition for closing the circle relaxed by a further 10^{-12} . Each point that it considers is the start of an arc plus $j\gamma_0$ for some $0 \leq j \leq 8$, so each of its comparisons is between $j\gamma_0$, $1 \leq j \leq 9$, and an integer multiple of the arc length $\pi/720$, and these always differ by more than $4 \cdot 10^{-5}$; every number compared arises from the start of the run, the arc length, $\gamma_0 - 10^{-12}$ and the previous point by a few floating-point operations on numbers of absolute value less than 14, and lies within 10^{-14} of the exact quantity it stands for. Hence every comparison has the same outcome as with the exact γ_0 and the closed arcs, and the computed count is the exact one. The re-verification takes from the search programs the grids of arcs and of radial intervals, the floating-point value of γ_0 , the floating-point test that proposes arcs to be discarded (each discard is then certified in Arb, so a rounding error there can only keep an arc), this final count and the bisection rule; the labels of the boxes therefore rest on this one implementation. A separate program checks that the accepted boxes leave no gap: it rebuilds the bisection tree from the initial boxes, accepting a node only if it coincides with a recorded box (each used exactly once) or contains no configuration with ordered angles, and confirms in exact rational arithmetic that the volumes of these boxes add up to that of the initial boxes (this checker reuses the bisection rule of the search, so it is not an independent implementation). The initial boxes and the radial range $[1, \Lambda]$ of the outer centres contain the parameter domain up to the arc $(\widetilde{2\pi}, 2\pi)$ just mentioned, since the floating-point values used for ρ_1 , Λ and $d + \frac{\sqrt{2}}{2}$ are upper bounds of the exact values (checked in high-precision arithmetic); in the re-verification the conditions themselves are evaluated with the exact constants, the floating-point values serving only as end points of the initial boxes. The programs, the lists of accepted boxes and the logs are part of the accompanying material, available at <https://github.com/squarepacker/k2-minus-c> and archived on Zenodo (<https://doi.org/10.5281/zenodo.23164301>, all versions); they were run with Python 3.12.10, `python-flint` 0.9.0, NumPy 2.5.3 and mpmath 1.4.1. That the accepted boxes leave no gap has also been confirmed by two further checks, written independently of the search programs and using different methods (exact rational volumes of the boxes intersected with the region of ordered angles, and a recursive covering of the domain); they were written by separate sessions of the same AI system that helped to write all the programs (see the section on the use of AI), and they share with the search the lists of boxes and the description of the domain. The verification of the individual boxes (the Arb re-verification) has not yet been reproduced by an independent implementation; the Squares Project reports a complete replay of it with these programs (from the release of version 1.0, whose programs and data for this lemma are identical), with the same results [21]; this repeats the same implementation. $\square$

Remark 4.11. The proof bounds the number of *all* pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p , without using the graph G , and for this number the bound 9 is sharp: the configuration described in Remark 7.5 below (two axis-parallel squares on both sides of a slit through p , each nearly touching two slightly tilted squares on either side) has 9 such pairs at an uncovered point; we checked this in 80-digit arithmetic for $d = 10^{-6}, 10^{-5}, 10^{-4}$ (and beyond the range of the lemma for $10^{-3}, 10^{-2}$), hence, by monotonicity in d , for every $d \in [10^{-6}, 10^{-4}]$ (the coordinates and the check are part of the accompanying material). We do not know whether 9 edges of a single graph G_h or G_v can meet in this way. This remark is not used in the proofs.

Apart from the definition of R_e , Lemma 4.8(b) is not used below; it bounds the multiplicity at all points, not only at uncovered ones.

Lemma 4.12 (slits). *Let A, B be distinct squares, or let one of them be the left or the right side of $[0, k]^2$, and let E_{AB} be the set of $y \in [0, k]$ such that on the line y both A and B are met, A lies to the left of B , and no square is met strictly between them. For $y \in E_{AB}$ let $g(y)$ be the length of the gap between them, and let $\theta := \theta(A, B)$.*

- (a) *All $y \in E_{AB}$ with $g(y) < 1$ lie in one component of E_{AB} .*
- (b) *On every component J of E_{AB} the function g is convex, and $\int_I g \geq \theta(2 - \theta)|I|^2/4$ for every interval $I \subset J$. In particular, for $\eta > 0$ the set $\{y \in J : g(y) < \eta\}$ is an interval.*

The same holds with the roles of the axes exchanged (vertical lines, A below B , the bottom and the top side).

Proof. Let U be the intersection of the vertical extents of A and B (for a side of the container, its vertical extent is $[0, k]$); it is an interval containing E_{AB} . Let $r_A(y)$ be the largest abscissa of A on the line y ($r_A \equiv 0$ for the left side) and $l_B(y)$ the smallest abscissa of B ($l_B \equiv k$ for the right side); by Lemma 4.2 (with $(s, t) = (x, y)$), $G := l_B - r_A$ is convex on U , and $g = G$ on E_{AB} . If a square Z lies strictly between A and B on one line $y \in U$, then it does so on every line meeting A, B and Z : two disjoint compact convex sets are strictly separated by a line, and if that line is not horizontal their order on every horizontal line meeting both is the same (if it is horizontal they never share a line); when one of A, B is a side of the container this is trivial, since every square lies to the right of the left side and to the left of the right side.

(a) Let $y_1 < y_2$ be points of E_{AB} with $g(y_1), g(y_2) < 1$; by convexity $G < 1$ on $[y_1, y_2]$. Suppose that some $y \in (y_1, y_2)$ is not in E_{AB} . Then a square Z is met strictly between A and B on the line y ; it does not meet the lines y_1 and y_2 , since it would lie between A and B there, so its vertical extent $V(Z)$ lies in $(y_1, y_2) \subset U$. The midpoint y_Z of $V(Z)$ lies in the middle band $\mathcal{M}(Z)$, so the chord of Z at height y_Z has length $\sec a(Z) \geq 1$ (Lemma 3.1(b)), and it lies strictly between A and B . Hence $G(y_Z) > 1$, a contradiction. So $[y_1, y_2] \subset E_{AB}$.

(b) On a component J of E_{AB} , $g = G$ is convex. By Lemma 4.2, it is piecewise linear with at most two break points, at each of which its slope increases by at least 2, and off the break points $|g'| = |\tan \psi_B - \tan \psi_A| \geq |\psi_B - \psi_A| \geq \theta$, where ψ_A, ψ_B are the angles between the vertical and the sides of A and B bounding the gap (for a side of the container the angle is 0, and $\theta(A, S_0)$ is the inclination of the square). As $\theta \leq \frac{\pi}{4} < 1$, Lemma 4.1 applies on every interval $I \subset J$. Sublevel sets of a convex function on an interval are intervals. $\square$

Lemma 4.13 (cost of a large inclination on a line). *Let $0 < \bar{\beta} \leq 1.5 \cdot 10^{-6}$, and let $Y \subset [0, k]$ and $\beta : Y \rightarrow [0, \bar{\beta}]$ be such that for every $y \in Y$: $\omega(y) < \frac{1}{2}$ and the line y meets a square of inclination $\geq \beta(y)$. Then*

$$\int_Y 2\beta(y) dy \leq AW, \quad A := \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}} < 10.6067.$$

Proof. For $y \in Y$ let $X_1, \dots, X_m$ be the squares met by the line, in order, and X_0, X_{m+1} the left and right sides of the container. Each gap is part of the line waste, so it is shorter than $\frac{1}{2}$, and the triangle inequality from the left side to a square of inclination $\geq \beta(y)$ and on to the right side gives $\sum_{j=0}^m \theta(X_j, X_{j+1}) \geq 2\beta(y)$. Put $\theta_{\max} := 2\bar{\beta}$ and $\theta' := \min(\theta, \theta_{\max})$; then also $\sum_j \theta'(X_j, X_{j+1}) \geq 2\beta(y)$, since either some term equals $\theta_{\max} \geq 2\beta(y)$, or no term is capped.

For a pair $e = \{P, Q\}$ of members of $\mathcal{V}_h$, with P the left one, let Y_e be the set of $y \in Y$ on which P and Q are consecutive, and let $\mathcal{E}$ be the finite set of pairs with $|Y_e| > 0$; they are edges of G_h . Write $\theta_e := \theta(P, Q)$, $\theta'_e := \min(\theta_e, \theta_{\max})$, $E_e := E_{PQ} \supset Y_e$, and g_e for the gap function of

Lemma 4.12. Integrating over Y ,

$$\int_Y 2\beta \leq \sum_{e \in \mathcal{E}} \theta'_e |Y_e|.$$

Put $d' := \theta_{\max}$ and split $Y_e = Y_e^c \cup Y_e^f$, where $Y_e^c := \{y \in Y_e : g_e(y) < d'\}$. Let $s_e^f := \int_{E_e \cap \{g_e \geq d'\}} g_e$. Since $\theta'_e \leq \theta_{\max} = d'$,

$$\theta'_e |Y_e^f| \leq \frac{\theta_{\max}}{d'} \int_{Y_e^f} g_e \leq s_e^f.$$

Now let $|Y_e^c| > 0$; then the members of e are at distance $< d' \leq 3 \cdot 10^{-6}$. As $g_e < \frac{1}{2}$ on Y_e , Lemma 4.12 shows that Y_e lies in one component J_e of E_e , and that Y_e^c lies in the interval $I_e := \{y \in J_e : g_e(y) < d'\}$, with

$$s_e^c := \int_{I_e} g_e \geq \frac{\theta_e(2 - \theta_e)}{4} |I_e|^2 \geq \frac{\theta'_e(2 - \theta_{\max})}{4} |I_e|^2,$$

because $x(2 - x)$ is increasing on $[0, 1]$. Let f_e be the area of the waste in R_e (notation of Lemma 4.8(b), with $d = d'$). We claim that $f_e \geq Q_* \theta'_e(2 - \theta_{\max})/4$. If $\theta_e \leq \theta_1 := 10^{-5}$, this follows from Proposition 4.5. Otherwise Lemma 4.8(a) gives $f_e \geq \theta_e(2 - \theta_e)/(4q_0) \geq \theta_1(2 - \theta_1)/40$, which exceeds $Q_* \theta_{\max}(2 - \theta_{\max})/4 \geq Q_* \theta'_e(2 - \theta_{\max})/4$ as $\theta_{\max} \leq 3 \cdot 10^{-6}$. (For a square X and a side, $R_e = R(X, S_0)$ may stand on another side of the container; this is harmless, as $\theta(X, S_0) = a(X)$.) Hence

$$\theta'_e |Y_e^c| \leq \theta'_e |I_e| = (\theta'_e |I_e|^2 \cdot \theta'_e)^{1/2} \leq \frac{4}{2 - \theta_{\max}} \left(\frac{s_e^c f_e}{Q_*} \right)^{1/2}.$$

The sets $\{(x, y) : y \in E_e, x \text{ in the gap of } e \text{ on the line } y\}$ are disjoint parts of the waste for different e , and I_e and $E_e \cap \{g_e \geq d'\}$ are disjoint, so $S^c + S^f \leq W$ for $S^c := \sum_e s_e^c$ and $S^f := \sum_e s_e^f$. By Lemma 4.10 (with $d = d'$), every point of the waste lies in R_e for at most K_w^* of the pairs with $|Y_e^c| > 0$, so $\sum f_e \leq K_w^* W$, the sum over these pairs. By the Cauchy–Schwarz inequality,

$$\sum_{e \in \mathcal{E}} \theta'_e |Y_e| \leq \frac{4\sqrt{K_w^*/Q_*}}{2 - \theta_{\max}} \sqrt{S^c W} + S^f \leq A \sqrt{(W - S^f)W} + S^f \leq AW,$$

the last step because $x \mapsto A\sqrt{(W - x)W} + x$ is decreasing on $[0, W]$ when $A \geq 2$. Finally $4\sqrt{9/0.32}/(2 - 3 \cdot 10^{-6}) < 10.6067$. $\square$

Lemma 4.14 (cost of a large inclination in a column). *Let $0 < \delta \leq 10^{-5}$ and $0 < \bar{\beta} \leq 3 \cdot 10^{-6}$. Let $X, X' \subset [0, k]$. For $x \in X$ let $\bar{y}_x \in (0, \frac{k}{2} - 1]$ and $\beta(x) \in [0, \bar{\beta}]$ be such that the segment $\sigma_x = \{x\} \times [0, \bar{y}_x]$ has vertical waste $< \delta$ and meets a square of inclination $\geq \beta(x)$; for $x \in X'$ let $\bar{y}'_x, \beta'(x)$ satisfy the same with the segment $\sigma'_x = \{x\} \times [k - \bar{y}'_x, k]$. Then*

$$\int_X \beta(x) dx + \int_{X'} \beta'(x) dx \leq AW.$$

Proof. For $x \in X$ list the squares met by σ_x from the bottom up to the first one of inclination $\geq \beta(x)$, preceded by the bottom side of the container; for $x \in X'$ list those met by σ'_x from the top down, preceded by the top side. All gaps are $< \delta$, so consecutive members are at distance $< \delta$, and the triangle inequality gives $\sum_j \theta(Y_j, Y_{j+1}) \geq \beta(x)$ (resp. $\beta'(x)$), hence the same for $\theta' := \min(\theta, \bar{\beta})$. Two members that are consecutive on σ_x are consecutive on the whole vertical line through x (a square between them would meet σ_x between them), so the pairs used for a set of x of positive measure are edges of G_v whose members are at distance $< \delta$.

Both members of a pair used for some $x \in X$ meet $\mathbb{R} \times [0, \frac{k}{2} - 1]$, and both members of a pair used for some $x \in X'$ meet $\mathbb{R} \times [\frac{k}{2} + 1, k]$; as no square meets both strips, no pair is used in both ways. For a pair e let X_e be the set of x in X , or in X' , at which it is used. Since the gaps are $< \delta < 1$, the vertical version of Lemma 4.12 puts X_e in one component J_e of E_e , and $s_e := \int_{J_e} g_e \geq \theta'_e(2 - \bar{\beta})|J_e|^2/4$. As in the proof of Lemma 4.13, with $d = \delta$ and $\theta_{\max}$ replaced by $\bar{\beta}$, we have $f_e \geq Q_*\theta'_e(2 - \bar{\beta})/4$ (by Proposition 4.5 if $\theta_e \leq \theta_1 = 10^{-5}$, and by Lemma 4.8(a) otherwise, since $\theta_1(2 - \theta_1)/40 \geq Q_*\bar{\beta}(2 - \bar{\beta})/4$). Hence $\theta'_e|X_e| \leq \theta'_e|J_e| \leq \frac{4}{2-\bar{\beta}}(s_e f_e/Q_*)^{1/2}$, $\sum_e s_e \leq W$ and $\sum_e f_e \leq K_w^* W$ (Lemma 4.10 with $d = \delta$), so the left-hand side is at most $\sum_e \theta'_e|X_e| \leq \frac{4\sqrt{K_w^*/Q_*}}{2-\bar{\beta}} W \leq AW$ (as $\bar{\beta} \leq 3 \cdot 10^{-6}$). $\square$

For $y \in (0, k)$ let $d(y) := \min(y, k - y)$ be the distance to the nearer of the bottom and the top side.

Lemma 4.15 (excess of long chords on a line). *Let $0 < \delta \leq 10^{-5}$ and $0 < \alpha \leq 1.5 \cdot 10^{-6}$, let $y \in (0, k)$ with $d(y) \leq \frac{k}{2} - 1$, and suppose that every square meeting the line y has inclination $< \alpha$. Then, with $E(y)$ as in Lemma 3.5,*

$$E(y) \leq 0.50001 \alpha \left(A + \frac{\alpha}{\delta} \right) W.$$

Proof. We use $\sec x - 1 \leq 0.50001 x^2$ for $0 \leq x \leq 10^{-3}$ (the function $(\sec x - 1)/x^2$ has a Taylor series with positive coefficients, so it is increasing on $(0, \frac{\pi}{2})$, and its value at 10^{-3} is less than 0.50001). Let S be a square with $c_S(y) \geq 1$, and $a := a(S) < \alpha$. By Lemma 3.1(b), $c_S(y) \leq \sec a$, so

$$c_S(y) - 1 \leq 0.50001 a^2 \leq 0.50001 \alpha a c_S(y),$$

as $a < \alpha$ and $c_S(y) \geq 1$. The chords of these squares on the line y are disjoint; let $U \subset [0, k]$ be the union of their projections on the x -axis and, for $x \in U$, let S_x be the square whose chord contains (x, y) . Summing,

$$E(y) \leq 0.50001 \alpha \sum_{S: c_S(y) \geq 1} a(S) c_S(y) = 0.50001 \alpha \int_U a(S_x) dx.$$

Let $y \leq \frac{k}{2}$; then $0 < y = d(y) \leq \frac{k}{2} - 1$. For $x \in U$ the segment $\sigma_x := \{x\} \times [0, y]$ meets S_x , of inclination $a(S_x) < \alpha$. Let U' be the set of $x \in U$ for which σ_x has vertical waste $\geq \delta$; as the vertical waste of σ_x is at most that of $\{x\} \times [0, k]$, whose integral over x is W , we have $|U'| \leq W/\delta$, and so $\int_{U'} a(S_x) dx \leq \alpha W/\delta$. Lemma 4.14, applied to $X := U \setminus U'$, $X' := \emptyset$, $\bar{y}_x := y$, $\beta(x) := a(S_x)$ and $\bar{\beta} := \alpha$, gives $\int_{U \setminus U'} a(S_x) dx \leq AW$. If $y > \frac{k}{2}$, the same argument with the segments $\{x\} \times [y, k]$ hanging from the top side (that is, $X := \emptyset$, $X' := U \setminus U'$ and $\bar{y}'_x := k - y = d(y)$) gives the same bound. Hence $\int_U a(S_x) dx \leq (A + \alpha/\delta)W$. $\square$

5. QUANTIZATION IN COLUMNS

Fix $C := 2$ and $\delta := 10^{-5}$, and put $\alpha(s) := 1/(Cs)$ for $s > 0$.

Definition 5.1. Let $0 < s \leq \frac{k}{2} - 3$. A point $x \in [0, k]$ is *s-rigid from below* if the segment $\sigma_x^s := \{x\} \times [0, s + 2]$ has vertical waste $< \delta$ and meets no square of inclination $\geq \alpha(s)$, and *s-rigid from above* if the segment $\tilde{\sigma}_x^s := \{x\} \times [k - s - 2, k]$ has these two properties. We write NR_s and $\widetilde{\text{NR}}_s$ for the sets of x that are not *s-rigid from below*, respectively from above.

If $s < s'$, then $\sigma_x^s \subset \sigma_x^{s'}$ and $\alpha(s) > \alpha(s')$, so $\text{NR}_s \subset \text{NR}_{s'}$. Moreover, for fixed x the set of $s \in (0, \frac{k}{2} - 3]$ with $x \in \text{NR}_s$ is closed: it is the union of the set where the vertical waste of σ_x^s , a continuous function of s , is $\geq \delta$, and of the sets $\{s : s + 2 \geq b_X, Cs a(X) \geq 1\}$ over the finitely

many squares X with $a(X) > 0$ that meet the vertical line through x , where b_X is the lower end point of the chord of X on that line. The same holds for $\widetilde{\text{NR}}_s$.

For a square Z and $x \in \mathcal{M}'(Z)$ let $\beta_Z(x)$ be the height of the lower endpoint of the vertical chord of Z at x . A square $X \neq Z$ is *below Z at x* if X meets $\{x\} \times [0, \beta_Z(x))$; its vertical chord at x is then contained in $[0, \beta_Z(x))$, because it is an interval disjoint from the chord of Z .

Lemma 5.2 (quantization). *Let $s \geq 2.36 \cdot 10^5$ with $s \leq \frac{k}{2} - 3$, let Z be a square with $a(Z) < \alpha(s)$ whose lowest point has height $v_Z \leq s$, and let $x \in \mathcal{M}'(Z)$ satisfy*

- (i) $x \in \mathcal{M}'(X)$ for every square X with $a(X) < \alpha(s)$ that is below Z at x , and
- (ii) x is s -rigid from below.

Then every point of $\mathcal{R}(Z)$ is at distance less than $w_0 := 1.265 \cdot 10^{-5}$ from an integer.

Proof. Write $\beta = \beta_Z(x)$ and $\alpha = \alpha(s) \leq 2.119 \cdot 10^{-6}$; by Lemma 3.1(c), $\beta \leq v_Z + \sin a(Z) < s + \alpha$, so $\{x\} \times [0, \beta) \subset \sigma_x^s$. By (ii) every square below Z at x has inclination $< \alpha$, so by (i) and Lemma 3.1(c) its vertical chord at x has length in $[1, \sec \alpha]$. If n is the number of these squares, their chords and the uncovered part fill $[0, \beta)$, whence $n \leq \beta < n \sec \alpha + \delta$ and $n < s + \alpha$. We use $\sec x - 1 \leq 0.50001 x^2$ for $0 \leq x \leq 10^{-3}$ (see the proof of Lemma 4.15). Now $n(\sec \alpha - 1) \leq (s + \alpha) \cdot 0.50001 \alpha^2 = 0.50001(1 + \alpha/s)/(C^2 s) < 5.3 \cdot 10^{-7}$, so $\beta \in [n, n + 1.053 \cdot 10^{-5})$. By Lemma 3.1(c), and as $\sin a(Z) < 2.119 \cdot 10^{-6}$, $v_Z \in [\beta - \sin a(Z), \beta] \subset (n - 2.119 \cdot 10^{-6}, n + 1.053 \cdot 10^{-5})$. The lower ramp of Z lies in $[v_Z, v_Z + \sin a(Z)] \subset (n - 2.119 \cdot 10^{-6}, n + 1.2649 \cdot 10^{-5})$. The upper ramp is $[v_Z + \cos a(Z), v_Z + \cos a(Z) + \sin a(Z)]$; here $v_Z + \cos a(Z) \geq \beta - \sin a(Z) + \cos a(Z) \geq n + \cos \alpha - \sin \alpha > n + 1 - 2.12 \cdot 10^{-6}$ and $v_Z + \cos a(Z) + \sin a(Z) \leq \beta + 1 + \sin a(Z) < n + 1 + 1.2649 \cdot 10^{-5}$. $\square$

For a square X with $a(X) > 0$ and horizontal extent $[x_L, x_R]$, the vertical ramp set of X (Lemma 3.1(c)) consists of two intervals of length $\sin a(X)$: the *left end* $[x_L, x_L + \sin a(X)]$ and the *right end* $[x_R - \sin a(X), x_R]$. They are the projections on the x -axis of the *left side* and the *right side* of X , the sides through the leftmost and the rightmost vertex that make the angle $a(X)$ with the vertical. If $a(X) = 0$ we say that the left end of X is the single point x_L (the abscissa of its left side).

Lemma 5.3 (cracks). *Let $0 < \alpha \leq 10^{-3}$ and $0 < \delta \leq 10^{-3}$ (in this lemma α and δ are arbitrary numbers in these ranges), and let $\sigma = \{x\} \times [0, b]$ be a vertical segment with vertical waste $< \delta$ such that every square meeting σ has inclination $< \alpha$. Let X be a square meeting σ such that x lies in the right end $[\xi_X, \xi_X + \sin a(X)]$ of X (so $0 < a(X) < \alpha$), let $p = (x, h)$ be the point where the vertical line through x meets the right side of X , and assume $h \leq b - \delta$. Then either $x - \xi_X < \delta \tan \alpha (1 + 2.01 \tan^2 \alpha)$, or there is a square $X' \neq X$ that meets σ at a point $q \neq p$ of the vertical segment of length δ that starts at p and leaves X there (so $q \notin X$ and $|q - p| < \delta$), and satisfies*

$$\lambda' \leq x \leq \lambda'' + 2.01 \delta \tan \alpha,$$

where $[\lambda', \lambda'']$ is the left end of X' .

Proof. Write $a := a(X)$, let ς be the right side of X , P_1 its end point at the rightmost vertex (abscissa $\xi_X + \sin a$), and P its other end point (abscissa ξ_X); let P_y denote the height of P . Then ς runs from P_1 in the direction $(-\sin a, u \cos a)$ for some $u \in \{1, -1\}$, and for heights y between those of P_1 and P the right end point of the chord of X at height y is $r(y) := x - u(y - h) \tan a$. Suppose that $x - \xi_X \geq \delta \tan \alpha (1 + 2.01 \tan^2 \alpha)$. Then $|P_y - h| = (x - \xi_X) \cot a \geq \delta(1 + 2.01 \tan^2 \alpha)$, so the heights $h + ut$ with $0 \leq t \leq \delta(1 + 2.01 \tan^2 \alpha)$ lie between those of p and P . Moreover the segment $\{x\} \times \{h + ut : 0 < t < \delta\}$ lies in σ : for $u = 1$ because $h \leq b - \delta$, for $u = -1$ because $P_y \geq 0$. As its uncovered part has length $< \delta$, it contains a point $q = (x, y_q)$, $y_q = h + ut$, $0 < t < \delta$, of some square X' . Since $r(y_q) = x - t \tan a < x$, the point q is not in X , so $X' \neq X$, and $a' := a(X') < \alpha$ because X' meets σ .

The disjoint convex sets X and X' both meet the horizontal line through q , so a line separating them is not horizontal, and on every horizontal line meeting both, X lies to the left of X' . Let $l'(y)$ be the left end point of the chord of X' at height y . Then (A) $l'(y) > r(y)$ for every height y between those of p and P at which X' is met, and (B) $x - t \tan a = r(y_q) < l'(y_q) \leq x$. Let $\hat{q} := (l'(y_q), y_q)$. If $a' = 0$, then $\hat{q}$ lies on the vertical left side of X' , and (B) gives the claim with $\lambda' = \lambda'' = l'(y_q)$. Let $a' > 0$, and let Q' be the leftmost vertex of X' , so that $\lambda' = Q'_x$. The left boundary $\{(l'(y), y)\}$ of X' consists of the left side, which projects onto $[\lambda', \lambda'']$, and of the other side F' of X' at Q' , which runs from Q' in the direction $(\cos a', v \sin a')$ for some $v \in \{1, -1\}$.

Case 1: $\hat{q}$ lies on the left side. Then $l'(y_q) \in [\lambda', \lambda'']$, and by (B) $\lambda' \leq x < \lambda'' + \delta \tan \alpha$.

Case 2: $\hat{q} = Q' + \mu(\cos a', v \sin a') \in F'$ with $0 < \mu \leq 1$, and $v = -u$. For $0 < w \leq \mu$ the point $\hat{q} - w(\cos a', v \sin a')$ of X' lies at height $y_q + uw \sin a'$. If this height lies between those of p and P , then (A) gives $l'(y_q) - w \cos a' > r(y_q + uw \sin a') = x - (t + w \sin a') \tan a$, that is, $w(\cos a' - \sin a' \tan a) < l'(y_q) - x + t \tan a \leq t \tan a$ by (B). Put $w_* := t \tan a / (\cos a' - \sin a' \tan a)$. If $\mu > w_*$, applying this to all $w \in (w_*, \mu]$ shows that $|P_y - y_q| \leq w_* \sin a'$, hence $|P_y - h| \leq t + w_* \sin a' < \delta(1 + 2.01 \tan^2 \alpha)$, contrary to the first paragraph. So $\mu \leq w_*$, and by (B) $x - \lambda' = (x - l'(y_q)) + \mu \cos a' < t \tan a + t \tan a / (1 - \tan a \tan a') < 2.01 \delta \tan \alpha$, while $x \geq l'(y_q) \geq \lambda'$.

Case 3: $\hat{q} = Q' + \mu(\cos a', v \sin a') \in F'$ with $0 < \mu \leq 1$, and $v = u$. Then the left side of X' runs from Q' in the direction $(\sin a', -u \cos a')$, and $Q'_y = y_q - u \mu \sin a'$, while $h = y_q - ut$. If $\mu \sin a' > t$, the height h lies strictly between Q'_y and y_q , so $l'(h) = l'(y_q) - t \cot a'$; by (A) and (B), $l'(h) > r(h) = x \geq l'(y_q)$, which is absurd. So $\mu \sin a' \leq t$; then the height h lies on the left side of X' , at distance $t - \mu \sin a' < \delta$ from Q'_y , so that $l'(h) \leq \lambda' + \delta \tan \alpha$, and (A) gives $x = r(h) < l'(h) \leq \lambda' + \delta \tan \alpha$; also $x \geq l'(y_q) \geq \lambda'$.

In all cases $\lambda' \leq x \leq \lambda'' + 2.01 \delta \tan \alpha$, and $q \in X' \cap \sigma$ is at distance $t < \delta$ from p . $\square$

Lemma 5.4 (room in a column). *Under the assumptions of Lemma 5.2 on s and Z , let $B_Z = B_Z^{(s)}$ be the set of points $x \in \mathcal{M}'(Z)$ that are s -rigid from below but violate (i). Then $|B_Z| < 0.500023$, so that $G_Z = G_Z^{(s)} := \mathcal{M}'(Z) \setminus B_Z$ has measure greater than $g_0 := 0.49997$.*

Proof. Write $\alpha = \alpha(s) \leq 2.119 \cdot 10^{-6}$, $\varrho := \cos \alpha + \sin \alpha$ and $\mathcal{M}'(Z) = [m_0, m_1]$, so that $m_1 - m_0 = \cos a(Z) - \sin a(Z) \leq 1$; note that the left end of Z ends at m_0 (if $a(Z) = 0$ it is the point m_0 ; if $a(Z) > 0$, then in the coordinates of the proof of Lemma 3.1 the horizontal extent of Z is $[x_0 - \sin a(Z), x_0 + \cos a(Z)]$, its left end is $[x_0 - \sin a(Z), x_0]$ and $\mathcal{M}'(Z) = [x_0, x_0 + \cos a(Z) - \sin a(Z)]$, and likewise in the reflected position). Let $\mathcal{X}_L$ be the set of squares X with $a(X) < \alpha$ that are below Z at some point of $\mathcal{M}'(Z)$ and whose left end is contained in (m_0, m_1) , and let $\mathcal{X}_R$ be the set of squares X with $0 < a(X) < \alpha$ that are below Z at some point of $\mathcal{M}'(Z)$ and whose right end is contained in (m_0, m_1) .

Counting. If $X \in \mathcal{X}_L$ and $y \in \mathcal{M}(X)$ (every y in the vertical extent of X if $a(X) = 0$), the chord of X at height y has its left end point in the left end of X , hence in (m_0, m_1) , and length $\geq 1 \geq m_1 - m_0$, so it contains m_1 . Chords of distinct squares at the same height are disjoint, so the middle bands of the squares in $\mathcal{X}_L$ have pairwise disjoint interiors. A square X below Z at some $x \in \mathcal{M}'(Z)$ meets $\{x\} \times [0, \beta_Z(x))$, and $\beta_Z(x) \leq v_Z + \sin a(Z) < s + \alpha$; so its middle band lies in $[0, s + \alpha + \varrho]$ and has length $\cos a(X) - \sin a(X) > \cos \alpha - \sin \alpha$. Hence $|\mathcal{X}_L| \leq n_* := (s + \alpha + \varrho) / (\cos \alpha - \sin \alpha)$, and in the same way (with m_0 in place of m_1) $|\mathcal{X}_R| \leq n_*$.

Covering. Let $x \in B_Z$. As (i) fails, some square X with $a(X) < \alpha$ is below Z at x and $x \notin \mathcal{M}'(X)$; then $a(X) > 0$ and x lies in the left or in the right end of X . If that end is not contained in (m_0, m_1) , it contains m_0 or m_1 , and $x \in [m_0, m_0 + \alpha) \cup (m_1 - \alpha, m_1]$. If it is the left end and it is contained in (m_0, m_1) , then $X \in \mathcal{X}_L$. If it is the right end and it is contained in (m_0, m_1) , then $X \in \mathcal{X}_R$, and we apply Lemma 5.3 to $\sigma = \sigma_x^s$ (here $b = s + 2$ and

$h < \beta_Z(x) < s + \alpha \leq b - \delta$; all squares meeting σ_x^s have inclination $< \alpha$, as x is rigid). Either $x \in [\xi_X, \xi_X + 1.01 \delta \tan \alpha]$, or there is a square X' as in the lemma; let q be the point of $X' \cap \sigma$ found there. If $X' = Z$, then $x \leq m_0 + 2.01 \delta \tan \alpha$. Otherwise $q \notin Z$, and since q lies within δ of p , which is below the chord $[\beta_Z(x), \beta_Z(x) + \sec a(Z)]$ of Z , the point q lies below that chord; so X' is below Z at x , and $a(X') < \alpha$. If the left end of X' is contained in (m_0, m_1) , then $X' \in \mathcal{X}_L$ and $x \in [\lambda', \lambda'' + 2.01 \delta \tan \alpha]$; otherwise $\lambda' \leq m_0$ or $\lambda'' \geq m_1$; in the first case $\lambda'' < m_0 + \alpha$ and $x < m_0 + \alpha + 2.01 \delta \tan \alpha$, and in the second case $x \geq \lambda' > \lambda'' - \alpha \geq m_1 - \alpha$ (if $a(X') = 0$, then $x \geq \lambda' = \lambda'' \geq m_1$). Altogether,

$$B_Z \subset [m_0, m_0 + 1.01\alpha] \cup [m_1 - \alpha, m_1] \cup \bigcup_{X \in \mathcal{X}_L} [\lambda'_X, \lambda''_X + 2.01 \delta \tan \alpha] \cup \bigcup_{X \in \mathcal{X}_R} [\xi_X, \xi_X + 1.01 \delta \tan \alpha].$$

As $\lambda''_X - \lambda'_X = \sin a(X) < \alpha$ and $\tan \alpha < 1.000001 \alpha$,

$$\begin{aligned} |B_Z| &\leq 2.01 \alpha + n_* \alpha (1 + 3.02 \delta \cdot 1.000001) \\ &\leq 2.01 \alpha + \frac{(1 + (\alpha + \varrho)/s)(1 + 3.04 \cdot 10^{-5})}{C(\cos \alpha - \sin \alpha)} < 0.500023, \end{aligned}$$

since $\alpha = 1/(Cs) \leq 2.119 \cdot 10^{-6}$, $s \geq 2.36 \cdot 10^5$, $\delta = 10^{-5}$ and $C = 2$ (the middle expression decreases as s grows); and $|\mathcal{M}'(Z)| = \cos a(Z) - \sin a(Z) > 1 - 2.12 \cdot 10^{-6}$. $\square$

Columns from the top. The reflection $(x, y) \mapsto (x, k - y)$ maps $[0, k]^2$ onto itself and closed packings to closed packings; since k is an integer, it maps points at distance less than w_0 from an integer to such points. Applying Lemmas 5.2–5.4 to the reflected packing, we obtain the same statements with “lowest point” replaced by “highest point”, $v_Z \leq s$ by $v'_Z \geq k - s$ for the height v'_Z of the highest point of Z , “below” by “above” (with the upper end point of the vertical chord of Z), and “rigid from below” by “rigid from above”; we denote the corresponding set by $\tilde{G}_Z^{(s)}$.

6. PROOF OF THEOREM 1.1

Put $\varepsilon := 2 \cdot 10^{-3}$, $\omega_0 := 2 \cdot 10^{-4}$, $y_{\min} := 236000$, $w_0 := 1.265 \cdot 10^{-5}$ and $h_0 := 1 - 2w_0$, and

$$F(k) := 1.99954 (\log k - 13.06675) / 30.418.$$

For $2 \leq k < 10^{12}$ we have $\log k < 27.632$, so $F(k) < 0.958 < 1 \leq k^2 - N$ by Corollary 3.3. Let now $k \geq 10^{12}$, and put $y_1 := \frac{k}{2} - 3$; recall $C = 2$, $\delta = 10^{-5}$ and $\alpha(s) = 1/(Cs)$ from Section 5, and put $C' := 2\sqrt{2}$ and $\alpha'(y) := 1/(C'y)$ (the threshold for lines). Recall that $d(y) = \min(y, k - y)$ is the distance to the nearer of the bottom and the top side, and for a measurable set $Y \subset \{y : d(y) \geq y_{\min}\}$ write $\ell(Y) := \int_Y dy/d(y)$. Let

$$H := \{y \in (0, k) : y_{\min} \leq d(y) \leq (1 - \varepsilon)y_1, \{y\} \in [w_0, 1 - w_0]\},$$

$H_b := H \cap (0, \frac{k}{2})$ and $H_t := H \cap (\frac{k}{2}, k)$. Since k is an integer, $y \mapsto k - y$ maps H_b onto H_t and preserves d , so $\ell(H) = 2\ell(H_b)$. If $j_0 := y_{\min} \leq m \leq j_1 - 1$ with $j_1 := \lfloor (1 - \varepsilon)y_1 \rfloor$ (note $j_1 > j_0$), then $[m + w_0, m + 1 - w_0] \subset H_b$ and $\int_{m+w_0}^{m+1-w_0} dy/y \geq h_0/(m+1) \geq h_0 \int_{m+1}^{m+2} dy/y$, so

$$\ell(H) \geq 2h_0 \log \frac{j_1 + 1}{j_0 + 1} \geq 2h_0 \log \frac{(1 - \varepsilon)(k/2 - 3)}{y_{\min} + 1} \geq 2h_0 (\log k - 13.06675),$$

the last step because $(1 - \varepsilon)(k/2 - 3) = \frac{1 - \varepsilon}{2} k(1 - 6/k)$, $\log \frac{2}{1 - \varepsilon} + \log(y_{\min} + 1) < 13.066741$ and, for $k \geq 10^{12}$, $\log(1 - 6/k) > -10^{-11}$.

For $y \in H$ we have $\alpha'(d(y)) \leq \alpha'(y_{\min}) < 1.4982 \cdot 10^{-6}$ and $d(y) \leq (1 - \varepsilon)y_1 < \frac{k}{2} - 1$. Every $y \in H$ is of exactly one of the following kinds: (i) $\omega(y) \geq \omega_0$; (ii) $\omega(y) < \omega_0$ and the line y

meets a square of inclination $\geq \alpha'(d(y))$; (iii) $\omega(y) < \omega_0$ and every square meeting the line y has inclination $< \alpha'(d(y))$. Let Y_i, Y_{ii}, Y_{iii} be the corresponding sets.

Kinds (i) and (ii). Since $\omega \geq \omega_0$ on Y_i and $d \geq y_{\min}$ on H , $W \geq \omega_0|Y_i| \geq \omega_0 y_{\min} \ell(Y_i)$. Lemma 4.13, applied to $Y = Y_{ii}$ with $\beta(y) = \alpha'(d(y)) \leq \bar{\beta} := 1.4982 \cdot 10^{-6}$, gives $AW \geq \int_{Y_{ii}} 2\alpha'(d(y)) dy = \frac{2}{C'} \ell(Y_{ii}) = \frac{1}{\sqrt{2}} \ell(Y_{ii})$.

Kind (iii): the excess of long chords. For $y \in Y_{iii}$, Lemma 4.15 with $\alpha = \alpha'(d(y))$ gives

$$E(y) \leq \gamma(y) W, \quad \gamma(y) := 0.50001 \alpha'(d(y)) \left(A + \frac{\alpha'(d(y))}{\delta} \right),$$

and, bounding the parts of H below and above $\frac{k}{2}$ separately (on both, $d(y) \geq y_{\min}$),

$$\begin{aligned} \Gamma &:= \int_H \gamma(y) \frac{dy}{d(y)} \leq 2 \cdot 0.50001 \int_{y_{\min}}^{\infty} \left(\frac{A}{C' y^2} + \frac{1}{C'^2 \delta y^3} \right) dy \\ &= 1.00002 \left(\frac{A}{C' y_{\min}} + \frac{1}{2C'^2 \delta y_{\min}^2} \right) < 1.61 \cdot 10^{-5}. \end{aligned}$$

Kind (iii): many non-rigid columns. Let $Y_b := Y_{iii} \cap H_b$ and $Y_t := Y_{iii} \cap H_t$. Let $s \in [y_{\min}, y_1]$ and $\mathcal{W}_s := [(1-\varepsilon)s, s]$, and let $\mathcal{Z}(s)$ be the set of squares Z with $\Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset$, where $\Phi(Z) = \{y : 0 < c_Z(y) < 1\}$ is as in Lemma 3.4. For $y \in Y_b \cap \mathcal{W}_s$ we have $d(y) = y$, and Lemma 3.5 gives $\sum_{Z: y \in \Phi(Z)} c_Z(y) \geq (1 - \omega_0 - E(y))_+$ (the sum is ≥ 0 , and $\omega(y) < \omega_0$); every Z in this sum belongs to $\mathcal{Z}(s)$, meets the line $y \leq s$ (so that the height v_Z of its lowest point satisfies $v_Z \leq s$), and has $0 < a(Z) < \alpha'(y) \leq \alpha'((1-\varepsilon)s)$ (by Lemma 3.4, $\Phi(Z) = \emptyset$ if $a(Z) = 0$). Integrating over $Y_b \cap \mathcal{W}_s$ and using $\int_{\Phi(Z)} c_Z = \sin a(Z) \cos a(Z) < \alpha'((1-\varepsilon)s)$ (Lemma 3.4), we get

$$|\mathcal{Z}(s)| \geq C'(1-\varepsilon)s \int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - E(y))_+ dy.$$

Let $Z \in \mathcal{Z}(s)$. As $C'(1-\varepsilon) > C$, we have $a(Z) < \alpha'((1-\varepsilon)s) < \alpha(s)$, so Lemmas 5.2 and 5.4 apply to s and Z . Every $x \in G_Z^{(s)}$ is either not s -rigid from below, or it is s -rigid from below and satisfies (i); in the latter case Lemma 5.2 would put every ramp point of Z within w_0 of an integer, which is impossible because Z has a ramp point $y \in H$ (as $\Phi(Z) \subset \mathcal{R}(Z)$). Hence $G_Z^{(s)} \subset \mathcal{M}'(Z) \cap \text{NR}_s$, and $|G_Z^{(s)}| > g_0$ by Lemma 5.4. Every $Z \in \mathcal{Z}(s)$ meets a line in $\mathcal{W}_s$ and has vertical extent < 1.0001 , and a vertical line through x meets every Z with $x \in G_Z^{(s)} \subset \mathcal{M}'(Z)$ in a chord of length ≥ 1 ; these chords are disjoint and lie in an interval of length $< \varepsilon s + 2.01$, so at most $\varepsilon s + 2.01$ of the sets $G_Z^{(s)}$ contain x . Hence

$$(1) \quad \mathcal{N}(s) := |\text{NR}_s| \geq \frac{g_0 |\mathcal{Z}(s)|}{\varepsilon s + 2.01} \geq \frac{g_0 C'(1-\varepsilon)s}{\varepsilon s + 2.01} \int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - E(y))_+ dy.$$

By the reflection $y \mapsto k - y$ and the versions of Lemmas 5.2 and 5.4 for columns from the top, $\tilde{\mathcal{N}}(s) := |\widetilde{\text{NR}}_s|$ satisfies (1) with $Y_b \cap \mathcal{W}_s$ replaced by $Y_t \cap (k - \mathcal{W}_s)$.

Kind (iii): the cost of non-rigid columns. For $x \in [0, k]$ let $T^*(x)$ be the least $s \in [y_{\min}, y_1]$ with $x \in \text{NR}_s$ (it exists by closedness, if $x \in \text{NR}_{y_1}$; otherwise $T^*(x)$ is undefined), and define $\tilde{T}^*(x)$ in the same way with $\widetilde{\text{NR}}_s$. By monotonicity, $\{x : T^*(x) \leq s\} = \text{NR}_s$ for $s \in [y_{\min}, y_1]$; as these sets are finite unions of intervals, T^* is a Borel function on the Borel set where it is defined, and $\{(x, s) : T^*(x) \leq s\}$ is a Borel set; likewise for $\tilde{T}^*$. If $T^*(x) = \tau$, then either (W) σ_x^τ has vertical waste $\geq \delta$, or (I) it has vertical waste $< \delta$ and meets a square of inclination $\geq \alpha(\tau)$; similarly for $\tilde{T}^*$ and $\tilde{\sigma}_x^\tau$. For each x the segments $\sigma_x^{T^*(x)} \subset \{x\} \times [0, \frac{k}{2} - 1]$ and $\tilde{\sigma}_x^{\tilde{T}^*(x)} \subset \{x\} \times [\frac{k}{2} + 1, k]$ are disjoint, so integrating the vertical waste over x shows that the x of type (W) for T^* and those

of type (W) for $\tilde{T}^*$ form two sets of total measure $\leq W/\delta$; on them $\alpha(T^*), \alpha(\tilde{T}^*) \leq \alpha(y_{\min})$. Lemma 4.14, applied to the x of type (I) with $\bar{y}_x = T^*(x) + 2$, $\beta(x) = \alpha(T^*(x))$, $\bar{y}'_x = \tilde{T}^*(x) + 2$, $\beta'(x) = \alpha(\tilde{T}^*(x))$ and $\bar{\beta} = \alpha(y_{\min}) < 2.119 \cdot 10^{-6}$, bounds the rest. Since $\alpha(\tau) \geq \int_{\tau}^{y_1} ds/(Cs^2)$, Tonelli's theorem gives

$$\left(A + \frac{\alpha(y_{\min})}{\delta}\right)W \geq \int \alpha(T^*(x)) dx + \int \alpha(\tilde{T}^*(x)) dx \geq \int_{y_{\min}}^{y_1} \frac{\mathcal{N}(s) + \tilde{\mathcal{N}}(s)}{Cs^2} ds.$$

For $y \in Y_b \subset [y_{\min}, (1-\varepsilon)y_1]$ the interval $[y, y/(1-\varepsilon)]$ lies in $[y_{\min}, y_1]$, and

$$\int_y^{y/(1-\varepsilon)} \frac{ds}{s(\varepsilon s + 2.01)} \geq \frac{1}{1 + 2.01/(\varepsilon y_{\min})} \int_y^{y/(1-\varepsilon)} \frac{ds}{\varepsilon s^2} = \frac{1}{(1 + 2.01/(\varepsilon y_{\min}))y}.$$

By (1) and Tonelli's theorem again, writing $\int_{Y_b \cap \mathcal{W}_s} f = \int_{Y_b} f(y) \mathbf{1}[y \leq s \leq y/(1-\varepsilon)] dy$, and in the same way for $\tilde{\mathcal{N}}$ (where $d(y) = k - y$),

$$\left(A + \frac{\alpha(y_{\min})}{\delta}\right)W \geq \frac{g_0 C'(1-\varepsilon)}{C(1 + 2.01/(\varepsilon y_{\min}))} \int_{Y_{iii}} (1 - \omega_0 - E(y))_+ \frac{dy}{d(y)}.$$

Conclusion. On Y_{iii} we have $1 - \omega_0 \leq (1 - \omega_0 - E)_+ + E \leq (1 - \omega_0 - E)_+ + \gamma W$, so $(1 - \omega_0)\ell(Y_{iii}) \leq \int_{Y_{iii}} (1 - \omega_0 - E)_+ dy/d(y) + \Gamma W$. Adding the bounds for the three kinds gives

$$\begin{aligned} (1 - \omega_0)\ell(H) &\leq \ell(Y_i) + \ell(Y_{ii}) + \int_{Y_{iii}} (1 - \omega_0 - E)_+ \frac{dy}{d(y)} + \Gamma W \\ &\leq \left(\frac{1}{\omega_0 y_{\min}} + \frac{C'}{2}A + \frac{C(1 + 2.01/(\varepsilon y_{\min}))}{C'g_0(1-\varepsilon)}\right)\left(A + \frac{\alpha(y_{\min})}{\delta}\right) \\ &\quad + \Gamma)W. \end{aligned}$$

With $y_{\min} = 236000$ (so $1/(\omega_0 y_{\min}) < 0.021187$, $2.01/(\varepsilon y_{\min}) < 4.2585 \cdot 10^{-3}$ and $\alpha(y_{\min})/\delta < 0.21187$), $A < 10.6067$, $g_0 = 0.49997$ and $\Gamma < 1.61 \cdot 10^{-5}$, the bracket is less than

$$\begin{aligned} &0.021187 + \sqrt{2} \cdot 10.6067 + \frac{1.0042585}{\sqrt{2} \cdot 0.49997 \cdot 0.998} (10.6067 + 0.21187) + 1.61 \cdot 10^{-5} \\ &= 30.41797 \dots < 30.418, \end{aligned}$$

so, as $(1 - \omega_0)h_0 \geq 0.99977$, $W \geq 2 \cdot 0.99977(\log k - 13.06675)/30.418 = F(k)$. Thus $W \geq \max(1, F(k))$ for every $k \geq 2$. If $0.0353 \log k \leq 1$, then $W \geq 0.0353 \log k$ by Corollary 3.3. Otherwise $\log k > 28.32$, and as $1.99954/30.418 - 0.0353 > 0.03043$ and $1.99954 \cdot 13.06675/30.418 < 0.85895 < 0.03043 \cdot 28.32$, we get $F(k) - 0.0353 \log k > 0.03043 \log k - 0.85895 > 0$. This proves Theorem 1.1 with $\kappa = 0.0353$. $\square$

7. REMARKS

Remark 7.1 (where integrality and closedness enter). Closedness is used in Lemma 3.2 (hence in Lemmas 3.4 and 3.5 and Corollary 3.3), in Propositions 4.3 and 4.5 and Lemmas 4.12 and 5.3 (separation of disjoint squares by a line), in Lemma 4.7 (consecutive squares do not touch), and in Lemma 2.1, which ties closed packings of $[0, k]^2$ to $s(n) < k$; for squares with disjoint interiors in a square of side exactly k , the $k \times k$ grid shows that nothing can be said. The disjointness of the squares is also used, in the arguments as given, in Lemmas 4.8, 4.9 and 4.10 (centres of disjoint squares are at distance at least 1; in Lemma 4.8(a) also the disjointness of Q_Z , S_1 and S_2), in Proposition 4.3(b) (the regions swept by the uncovered parts are disjoint parts of the waste), in Lemma 4.12(a) (the strict inequality $G(y_Z) > 1$), in Lemmas 4.13 and 4.14 (the gap regions of different pairs are disjoint parts of the waste), in Lemmas 4.15 and 5.4 (chords

of distinct squares at one height are disjoint; in the latter also $q \notin Z$), in Lemma 5.2 and the definition before it (vertical chords of distinct squares at one abscissa are disjoint), and in Section 6 (disjoint vertical chords bound the multiplicity of the sets $G_Z^{(s)}$). The integrality of the side enters through Lemma 3.2 (the bound $k - 1$ on long chords) and through the columns hanging from the top side: the reflection $y \mapsto k - y$ preserves the heights close to integers only because k is an integer. The quantization of Lemma 5.2 itself only uses one side, the unit size of the squares and their disjointness.

Remark 7.2 (large inclinations). The same line argument without the column analysis gives the following dichotomy. Let $k \geq 3 \cdot 10^{10}$ and $\alpha := 1/(4\sqrt{k}) \leq 1.45 \cdot 10^{-6}$. The lines with $\omega(y) \geq \frac{1}{2}$ have measure at most $2W$; by Lemma 4.13 with $\beta \equiv \alpha$, the lines with $\omega(y) < \frac{1}{2}$ that meet a square of inclination $\geq \alpha$ have measure at most $AW/(2\alpha)$; and on the remaining lines Lemma 3.4 gives short chords of total length $> 1 - \omega(y) - 0.0313$. Integrating,

$$0.9687k \leq (2.94 + 1.94A\sqrt{k})W + \sum_{0 < a(S) < \alpha} \sin a(S) \cos a(S).$$

Thus a closed packing with $W = o(\sqrt{k})$ must contain at least $(3.87 - o(1))k^{3/2}$ squares of inclination in $(0, \alpha)$. This remark is not used in the proofs.

Remark 7.3 (the threshold and the constant as functions of k). Put $F(k) := 1.99954(\log k - 13.06675)/30.418$, so that every closed packing satisfies $W = k^2 - N \geq \max(1, F(k))$ for every $k \geq 2$ (Section 6), and let $\kappa(k_1) := \inf_{k \geq k_1} \max(1, F(k))/\log k$ be the largest constant that these bounds certify for all $k \geq k_1$. It is non-decreasing in k_1 . As the additive constant is negative, $F(k)/\log k = \frac{1.99954}{30.418}(1 - 13.06675/\log k)$ increases with k , towards its limit $1.99954/30.418 = 0.065735\dots$, and $F(k)$ exceeds 1 only for $\log k > 28.2792\dots$; below this point the certified ratio is $1/\log k$, and beyond it $F(k)/\log k$, which there exceeds $1/28.2793$. Hence $\kappa(k_1) = 0.035361\dots$ for $2 \leq k_1 \leq e^{28.279}$, and $\kappa(k_1)$ increases to $0.065735\dots$ as $k_1 \rightarrow \infty$. In contrast with versions 1.0 and 1.1 of this paper, the constant for all k is thus limited not by the slope but by the point at which $F(k)$ exceeds 1, that is, by the additive constant $13.06675 \approx \log(2(y_{\min} + 1)/(1 - \varepsilon))$ coming from the smallest scale $y_{\min}$. A larger $y_{\min}$ slightly increases the slope but decreases the constant for all k (with unrounded constants, $y_{\min} = 3 \cdot 10^5$ and 10^6 give slopes 0.0659 and 0.0663 and constants 0.0351 and 0.0338 for all k , all rounded down), while with Lemma 4.13 as stated $y_{\min}$ cannot be smaller than $1/(C' \cdot 1.5 \cdot 10^{-6}) = 235702.26\dots$ (and Lemmas 5.2 and 5.4 are stated for $s \geq 2.36 \cdot 10^5$; at $s = 235702.26$ the value $w_0 = 1.265 \cdot 10^{-5}$ would not suffice). The limit $1/(\sqrt{2}A_\infty) = \sqrt{Q_*/K_w^*}/(2\sqrt{2}) \approx 0.066667$, where $A_\infty = 2\sqrt{K_w^*/Q_*}$, is the most that this architecture can give for the slope with $K_w^* = 9$ and $Q_* = 0.32$. For comparison, versions 1.0 and 1.1 of this paper, which used only the lines with $d(y) \geq \sqrt{k}/4$, had the bound $F(k) = (0.99977 \log k + 0.3819)/30.147$ for $k \geq 10^{13}$ and gave $\kappa = 0.033$ (its ratio $F(k)/\log k$ decreased to the limit $0.033163\dots$), the bound $F(k) = (0.99996 \log k + 0.386)/36.081$ of an earlier draft (with $K_w = 13$ and the threshold 10^{16}) gave $\kappa = 0.027$, the bound $F(k) = (0.99996 \log k + 0.386)/37.442$ of an earlier draft (with $K_w = 14$) gave $\kappa = 0.026$, the bound $F(k) = (0.999996 \log k + 0.386)/51.963$ of an earlier draft (with the multiplicity $K = 27$ in place of K_w) exceeded 1 only for $\log k > 51.58$ and gave $\kappa = 0.019$, and the bound $F(k) = (0.9987(\log k - 1.3884) - 0.0313)/84.05$ of an earlier draft exceeded 1 only for $\log k > 85.58$, and the constant it gave for all $k \geq 2$ was $1/85.58 \approx 0.0117$. Constant lower bounds for W from other sources raise the constant for all k without changing the slope: with the conditional bound $W \geq 4$ for $k \geq 6$ of Corollary 1.3 the infimum becomes $4/73.91674\dots = 0.054114\dots$

Remark 7.4 (the rate). The proof gives $c^*(k) \gg \log k$, while the upper bounds give $c^*(k) = O(k^{3/5})$. Several heuristics (balancing the cost of inclinations against the quantization near

the walls, and numerical linear-programming bounds for periodic covering measures) suggest that the truth may be of order $\sqrt{k}$; we leave this open. The constants in Theorem 1.1 are far from useful for concrete values: since W is an integer, the second bound of the theorem yields $s(k^2 - c) = k$ for every $c \geq 1$ as soon as $\log k > 15.22c + 13.07$; for instance (with the exact threshold $13.06675 + 30.418c/1.99954$) it yields $s(k^2 - 4) = k$ for $\log k > 73.92$ (compare the computer-assisted, unrefereed announcement of $s(k^2 - 4) = k$ for all $k \geq 5$ in [10]), and it says nothing about $s(12)$ or about any k within computational reach.

Remark 7.5 (the constant). An earlier draft of this argument, which used the lemma of [25] with its stated constant 10^{-10} and coarser bookkeeping, obtained only $\kappa \approx 6 \cdot 10^{-17}$, and previous drafts of this paper obtained $\kappa = 3 \cdot 10^{-4}$, $1.9 \cdot 10^{-3}$, 0.011, 0.019, 0.026, 0.027 and (versions 1.0 and 1.1) 0.033. The present constant comes from Proposition 4.3 (convex gaps, and cuts only where the set of squares met by the chords changes) and its refinement Proposition 4.5 for nearly parallel pairs, which replaces the factor $1/q_0 = 1/10$ by $Q_* = 0.32$; from the angular argument of Lemma 4.9 and its computer-assisted refinement Lemma 4.10, which bound the multiplicity of the rectangles at uncovered points, the only ones that matter, by $K_w = 13$ and $K_w^* = 9$, the latter sharp for all pairs at distance $< d$ (the planarity argument of Lemma 4.8(b), with Fodor's theorem [13] and Oler's inequality [24, 14], gives $K = 27$ at every point and is no longer needed); from Lemma 4.12(a), by which each pair is charged to a single slit; from capping the inclinations and splitting off the lines with a wide gap in Lemma 4.13; from Lemmas 3.4 and 3.5, by which the short chords on a line with little waste add up to almost a full unit (this halves the cost of the columns); from Lemma 4.15, by which the excess of the long chords is paid for by the waste, so that the lines down to the constant distance $y_{\min}$ from the walls can be used (this doubles the logarithmic range of scales once more, and with it the slope); from Lemma 5.3, by which a rigid column through the end of a square passes through a tight crack, so that the exceptional columns under a square have measure about $1/C$ rather than $2/C$; from measuring heights from both the bottom and the top side, which doubles the logarithmic range of scales; and from working with a continuum of scales instead of dyadic bands. With $g_0 \approx 1 - \frac{1}{C}$ the bracket in the proof is about $(\frac{C'}{2} + \frac{C}{C'(1-1/C)})A$, which is minimal, equal to $2\sqrt{2}A$, at $C = 2$, $C' = 2\sqrt{2}$. The factor $\sqrt{K_w^*/Q_*} \approx 5.3$ in A comes from using Propositions 4.3 and 4.5 for pairs that are consecutive only on a short set of lines, typically at corners where several squares meet, whereas in simple configurations, such as a single slightly tilted square in a grid, the ratio of $\int_Y 2\beta$ to W in Lemma 4.13 is of order 1. A lemma charging such short-lived pairs to the slits of neighbouring long-lived pairs with a bounded constant would improve κ further; we have not been able to prove one. In numerical experiments (randomized dense packings and targeted searches) the largest multiplicity of the rectangles R_e found by the accompanying search programs was 5 (against 27; an earlier search, whose program was not preserved, reported 8); and the largest number of pairs of squares at distance $< 10^{-4}$ (whether or not they are edges of G_h or G_v) whose rectangles $R(P, Q)$ contain a common uncovered point was 9, equal to K_w^* , in a degenerate configuration: two axis-parallel squares on both sides of a slit of width about 10^{-20} through the point, each nearly touching two slightly tilted squares on either side (random searches found at most 6). We also found no configuration with $\sum \lambda_r^2 < \frac{1}{3}$ in Proposition 4.5 (against 0.32; the smallest value found was 0.333335). The search programs and their results are part of the accompanying material. With $Q_* = \frac{1}{3}$ the same proof (with Lemma 4.13 stated for $\bar{\beta} \leq 1.4999 \cdot 10^{-6}$ and Lemma 4.14 for $\bar{\beta} \leq 2.99 \cdot 10^{-6}$, which still cover their uses in Section 6) would give the slope 0.0670 and the constant 0.0357 for all k (both rounded down). Only the numerical constants depend on the computer-assisted Lemma 4.10: with the analytic bound $K_w = 13$ of Lemma 4.9 in place of K_w^* , the proof of Section 6 gives $A < 12.7476$, a bracket less than 36.493 (with $\Gamma < 1.93 \cdot 10^{-5}$), the slope $1.99954/36.493 > 0.0547$ and $\kappa = 0.0319$. Even

with A of order 1, this architecture would limit the slope to roughly $\frac{1}{\sqrt{2}A}$, and, with the present $y_{\min}$, the constant for all k to less than $1/13.06$, so that a bound useful for concrete k would require a different rate rather than better constants.

Changes from version 1.1. The lines near the walls are now used. Lemma 3.5 (short chords against the excess $E(y)$ of the long chords) and Lemma 4.15 (a bound of $E(y)$ by the waste on lines whose squares have small inclination) are new; in Section 6 the cutoff $d(y) \geq \sqrt{k}/4$ is replaced by the constant $d(y) \geq y_{\min} = 236000$, the term $\eta(y)$ is replaced by $E(y)$, and the resulting integral Γ enters the bracket; the second bound of the theorem now holds for every $k \geq 2$ (for $k < 10^{12}$ it follows from Corollary 3.3), so the threshold $k_2 = 10^{13}$ is no longer needed. Lemma 4.14 is stated for $\bar{\beta} \leq 3 \cdot 10^{-6}$ (its proof is unchanged), and Lemmas 5.2 and 5.4 for $s \geq 2.36 \cdot 10^5$, with $w_0 = 1.265 \cdot 10^{-5}$ and $g_0 = 0.49997$. The constant improves from $\kappa = 0.033$ to $\kappa = 0.0353$, and the asymptotic slope from 0.0331 to 0.0657 ; the numbers derived from them (the threshold for $s(k^2 - 4) = k$, the constant without Lemma 4.10, and Remarks 7.3 and 7.5) are updated accordingly. Corollary 1.3 is new: it records the constant 0.0541 for all $k \geq 2$ that follows if one also uses the unrefereed computer-assisted result $s(k^2 - 3) = k$ ($k \geq 6$) of [10]; the references [20, 26] are new, and the entry [10] lists the further files used. Following the findings of an AI review of version 1.1 by the Squares Project [21], the following passages were also revised, without changing the results: the standing assumption of Section 4 and the proof of Lemma 4.10 now agree that $k \geq 4$ there, and Section 4 states that Sections 4 and 5 are used in the proof of Theorem 1.1 only for $k \geq 10^{12}$; the proof of Lemma 4.10 now explains why the extension of the angle intervals to 2π covers the arc $(\widetilde{2\pi}, 2\pi)$, shows that the floating-point count of points is exact, says which parts of the search programs the re-verification uses, describes the authorship of the two further coverage checks, and reports the replay of the computation by the Squares Project; the first remark of Section 7 lists further uses of disjointness; and the remark after Lemma 4.10 and the remark on large inclinations state that they are not used in the proofs. The new reference [21] records that review. Nothing else in the mathematics has changed.

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Use of AI. Claude (Anthropic) was used extensively in developing the arguments, writing the text and the programs (including the computation for Lemma 4.10), and in checking them. It is not an author; the author takes full responsibility for the contents. The paper has not been refereed.

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